CLIMB-Flow: derivation of the stage-wise sampler

Abstract

We derive a sampler for the stage-wise target arising from a cascaded pixel-domain flow prior in linear inverse problems. A change of variables makes one of the two conditionals of the target Gaussian, so it can be drawn exactly by a single conjugate-gradient solve with no step size. The other conditional is not Gaussian and is handled by a Langevin step, whose score is obtained in closed form from one network evaluation. Nothing is stated below that Algorithm 2 does not use. All proofs are collected in Appendix A.

1 Setup

1.1 The cascaded flow model

Let $D^{(1)}$ and $U^{(1)}$ be one-step linear downsampling and upsampling, and $G := U^{(1)} \circ D^{(1)}$. We assume G is symmetric with $0 \preceq G \preceq I$, which holds for the standard interpolation kernels. The index $k \in \{0, \ldots, K-1\}$ runs over resolution stages, coarsest to finest, and $\tau \in [0,1]$ is the time within a stage. The stage boundaries (s_k, e_k) with $0 \le s_k < e_k \le 1$ are fixed by the pretrained model.

At stage k the clean endpoint is $x_1^k = D^{K-1-k} x_1$ and the noise endpoint is $x_0^k \sim \mathcal{N}(0, I)$, independent of x_1^k. The two stage boundaries share the same noise endpoint:

$$x_{\text{start}}^k = s_k\, G\, x_1^k + (1-s_k)\, x_0^k, \qquad x_{\text{end}}^k = e_k\, x_1^k + (1-e_k)\, x_0^k. \tag{1}$$

The intermediate state $x_\tau^k = (1-\tau)x_{\text{start}}^k + \tau x_{\text{end}}^k$ is therefore linear in the pair, with Gaussian noise:

$$\underbrace{x_\tau^k}_{\text{interpolant}} = \underbrace{H_\tau^k}_{(1-\tau)s_k G + \tau e_k I} x_1^k + \underbrace{\sigma_\tau^k}_{(1-\tau)(1-s_k) + \tau(1-e_k)} x_0^k. \tag{2}$$

Subtracting the two relations of (1) gives the stage displacement,

$$x_{\text{end}}^k - x_{\text{start}}^k = \underbrace{B_k}_{e_k I - s_k G} x_1^k - (e_k - s_k)\, x_0^k. \tag{3}$$

Both H_τ^k and B_k are polynomials in G, hence symmetric, and they commute. The same holds for the operator N_k introduced in Prop. 7. Consequently

$$(H_\tau^k)^\top H_\tau^k = (H_\tau^k)^2, \qquad N_k^\top N_k = N_k^2, \tag{4}$$

and we use the two notations interchangeably: wherever a square of H_τ^k or N_k appears below it denotes the Gram matrix, and the transpose has been dropped only because these operators are self-adjoint. The forward operator A_k is *not* symmetric, so $A_k^\top A_k$ is always written with its transpose. Everything that follows uses (4), so an implementation in which G fails to be symmetric — through boundary handling, for instance — will violate the derivations silently; Section 7 makes this the first thing to test. We write p^k_τ for the law of x^k_τ. The velocity network is evaluated at the interpolant and estimates the conditional mean of the displacement,

$$v_\theta(x^k_\tau, \tau, k) \;\approx\; d^k_\tau := \mathbb{E}\big[x^k_{\text{end}} - x^k_{\text{start}} \,\big|\, x^k_\tau\big]. \tag{5}$$

Equivalently, v_θ supplies reconstructions of the two stage boundaries,

$$\widehat{x}^k_{\text{start}} = x^k_\tau - \tau\, v_\theta, \qquad \widehat{x}^k_{\text{end}} = x^k_\tau + (1-\tau)\, v_\theta, \tag{6}$$

and conversely $v_\theta = \widehat{x}^k_{\text{end}} - \widehat{x}^k_{\text{start}}$ and $x^k_\tau = (1-\tau)\widehat{x}^k_{\text{start}} + \tau \widehat{x}^k_{\text{end}}$.

1.2 The inverse problem

The measurement model is $y = Ax_1 + n$ with $n \sim \mathcal{N}(0, \eta^2 I)$ and A a known linear operator. At stage k we use $A_k := A \circ U^{K-1-k}$, so that

$$y = A_k\, x^k_1 + n, \qquad p(y \mid x^k_1) \;\propto\; \exp\Big(-\tfrac{1}{2\eta^2}\|A_k x^k_1 - y\|^2\Big). \tag{7}$$

Here n is the dimension of x^k_1 and m the dimension of y. Table 1 collects the symbols.

Table 1: Symbols. The sampler state is the pair (x^k_1, x^k_0); everything else is data, fixed by the stage schedule, or derived from the state.

Symbol	Definition	Role
x^k_1	$D^{K-1-k}x_1$, clean image at scale k	**sampled**
x^k_0	noise endpoint, $x^k_0 \sim \mathcal{N}(0, I)$	**sampled**
x^k_τ	$H^k_\tau x^k_1 + \sigma^k_\tau x^k_0$, see (2)	derived from the state
y	$A_k x^k_1 + n$, see (7)	data, fixed
H^k_τ, σ^k_τ	see (2)	fixed given (k, τ)
B_k	$e_k I - s_k G$, see (3)	fixed given k; independent of τ
A_k, η	see (7)	fixed given k
v_θ	network output at (x^k_τ, τ, k)	computed, one call per iteration
d^k_τ	conditional mean of the displacement	estimated by v_θ
γ^2	variance of the velocity error	sets the ridge in (38)
h_0	Langevin step size for x^k_0	the only tuned step size

1.3 Why the prior must be blurred

Bayes' rule for the model of Section 1 gives the posterior

$$p(x^k_1 \mid y) \;\propto\; \underbrace{p(y \mid x^k_1)}_{\text{likelihood, (7)}} \; \underbrace{p^k_1(x^k_1)}_{\text{prior on the } \textit{clean} \text{ image}}, \tag{8}$$

which cannot be sampled as it stands, for two separate reasons.

The prior is degenerate. Under the manifold hypothesis p^k_1 is supported on a set of measure zero in $\mathbb{R}^n$. Then $\log p^k_1 = -\infty$ off that set, and $\nabla \log p^k_1$ is undefined there. A sampler initialized away from the support has no prior gradient to follow, and (8) inherits the degeneracy, since the likelihood is positive everywhere and does not repair it. Replacing p^k_1 by a Gaussian-smoothed version makes the density strictly positive and smooth on all of $\mathbb{R}^n$, so the score exists everywhere and the smoothed posterior has full support. This is the standard reason smoothed priors appear in samplers built on diffusion and flow models, and it is not particular to the present construction.

The prior is unavailable. Even where $\nabla \log p_1^k$ exists, it is not what the pretrained network provides. The network represents p_τ^k, the law of the interpolant, and only through its score.

Both difficulties are met by working with a family of blurred priors indexed by the stage schedule.

The family. Let $\tau > 0$, so that $H_\tau^k = (1-\tau)s_k G + \tau e_k I$ is invertible. Write $X \sim p_1^k$ for the clean image and $Z \sim \mathcal{N}(0, I)$ for a standard Gaussian independent of it. In place of $p_1^k(x_1^k)$ we take the density $\widetilde{p}_\tau^k(x_1^k)$ of the blurred variable,

$$\widetilde{p}_\tau^k(x_1^k) \;:=\; \text{density at } x_1^k \text{ of } \quad \underbrace{X}_{\sim\, p_1^k} \;+\; \underbrace{\sqrt{2}\,\sigma_\tau^k\,(H_\tau^k)^{-1} Z}_{\text{blur, covariance } 2(\sigma_\tau^k)^2 (H_\tau^k)^{-2}} \qquad . \tag{9}$$

Every member of the family is smooth and strictly positive. The blur vanishes as $\sigma_\tau^k \to 0$ and $H_\tau^k \to I$, and both hold at the terminal step of the final stage, where $\tau = 1$ and $e_{K-1} = 1$; there $\widetilde{p}_\tau^k = p_1^k$. The family is therefore an annealing: coarse stages target a heavily blurred prior, the final step targets the true one. Lemma 1 shows that $\widetilde{p}_\tau^k$ is the law of the variable constructed there, which is why that variable is named X_1^k.

On the covariance in (9). The magnitude and the shape of the blur have different status, and it is worth separating them.

The magnitude $2(\sigma_\tau^k)^2$ is fixed by the schedule and is not chosen. It is $\sqrt{2}\,\sigma_\tau^k$ rather than σ_τ^k because two independent units of noise at scale σ_τ^k are combined, as Step 2 of Lemma 1 shows; it decreases monotonically along the cascade.

The shape $(H_\tau^k)^{-2}$ is an artifact of the coordinates, not a modelling decision. In the space in which the network operates, that of the interpolant, the added noise is $\sqrt{2}\,\sigma_\tau^k Z$, with covariance $2(\sigma_\tau^k)^2 I$: it is *isotropic*. The colouring appears only on pulling back to image space through $(H_\tau^k)^{-1}$, which is the map (12) uses to return from interpolant coordinates. A plain isotropic blur in image space would serve the purpose stated above equally well; the coloured form is simply what this model's geometry produces, and it introduces no parameter to tune.

The colouring is nonetheless interpretable. Since $(H_\tau^k)^{-2}$ is large where H_τ^k is small, and $H_\tau^k = (1-\tau)s_k G + \tau e_k I$ is smallest on $\ker(G)$ and at small τ, the prior is blurred most in the directions the interpolant attenuates most, which are exactly the directions in which x_τ^k carries least information about x_1^k. Being least committal there is reasonable, though nothing below relies on it. Whether an isotropic blur of the same magnitude performs comparably is an empirical question we do not settle.

1.4 The posterior we target

Replacing p_1^k by $\widetilde{p}_\tau^k$ in (8) gives the object we actually want to sample,

$$\pi(x_1^k \mid y) \;\propto\; \underbrace{p(y \mid x_1^k)}_{\text{likelihood}} \quad \underbrace{\widetilde{p}_\tau^k(x_1^k)}_{\text{blurred prior, (9)}} \qquad . \tag{10}$$

This is an exact posterior; the blurring of (9) is the only approximation made in this section.

Equation (10) cannot be used directly, because (9) defines $\widetilde{p}_\tau^k$ through p_1^k, which is unavailable. The remedy is to introduce the noise endpoint as a second variable. Lemma 1 exhibits $\widetilde{p}_\tau^k$ as the marginal of a two-variable model whose conditional is expressed through p_τ^k, and Bayes' rule applied to that model turns (10) into the joint target (16) that the sampler uses.

1.5 The augmented target

Step 1: the blurred prior is a marginal of a two-variable model. Equation (9) is stated in terms of p_1^k, which is unavailable. Lemma 1 re-expresses it in terms of p_τ^k, which is available, at the cost of introducing the noise endpoint as a second variable.

Lemma 1 (The blurred prior, expressed through p_τ^k)**.** *Let $X \sim p_1^k$ and $Z_1 \sim \mathcal{N}(0, I)$ be independent. By (2), p_τ^k is the law of*

$$X_\tau^k \;=\; H_\tau^k\, X + \sigma_\tau^k\, Z_1. \tag{11}$$

Let $X_0^k \sim \mathcal{N}(0, I)$ be independent of X and Z_1, and define

$$X_1^k \;:=\; (H_\tau^k)^{-1}\big(X_\tau^k - \sigma_\tau^k\, X_0^k\big). \tag{12}$$

Then (a) *the density of X_1^k is $\widetilde{p}_\tau^k$ of (9), so (12) realizes Assumption (A); and* (b) *the conditional density of X_1^k given $X_0^k = x_0^k$ is*

$$p(x_1^k \mid x_0^k) \;=\; |\det H_\tau^k|\; p_\tau^k\big(H_\tau^k x_1^k + \sigma_\tau^k x_0^k\big). \tag{13}$$

Is the blurred prior, or its score, computable? Its *value* is not: by Step b1 it is an average of p_τ^k over noisy points, and p_τ^k is available only through its score. Its *score* is available, in the following sense. Write $q(x_1^k, x_0^k) \propto p_\tau^k(H_\tau^k x_1^k + \sigma_\tau^k x_0^k)\,\mathcal{N}(x_0^k; 0, I)$ for the joint of Lemma 1, whose x_1^k-marginal is $\widetilde{p}_\tau^k$. Differentiating that marginal and using $\nabla_{x_1^k} \log q = (H_\tau^k)^\top \nabla \log p_\tau^k(x_\tau^k)$,

$$\nabla_{x_1^k} \log \widetilde{p}_\tau^k(x_1^k) \;=\; \mathbb{E}_{x_0^k \sim q(\cdot \mid x_1^k)}\Big[\,\underbrace{(H_\tau^k)^\top \nabla \log p_\tau^k\big(H_\tau^k x_1^k + \sigma_\tau^k x_0^k\big)}_{\text{network score at a noisy point}}\,\Big], \tag{14}$$

which is Fisher's identity: the score of a marginal is the conditional average of the score of the joint. So the blurred prior's score is an average of network evaluations at points x_τ^k of exactly the kind the model was trained on — one adds noise to x_1^k through x_0^k and evaluates there.

Two consequences. Computing (14) directly would require drawing x_0^k from $q(\cdot \mid x_1^k)$ at every x_1^k, an inner loop. The augmented sampler avoids it: x_0^k is carried in the state, and Block 2 is precisely a kernel for $q(\cdot \mid x_1^k)$, so the average is performed by the chain rather than by an explicit estimator. And a single draw of x_0^k already gives an unbiased estimate of (14), which is why one network evaluation per iteration suffices.

Step 2: Bayes' rule. Lemma 1 exhibits the blurred prior as the chain $X_0^k \to X_1^k \to Y$, in which Y depends on X_0^k only through X_1^k because (7) involves x_1^k alone. Bayes' rule for that chain reads

$$\pi(x_1^k, x_0^k \mid y) \;\propto\; \underbrace{p(y \mid x_1^k)}_{\text{likelihood, (7)}} \;\cdot\; \underbrace{p(x_1^k \mid x_0^k)}_{\text{transition, (13)}} \;\cdot\; \underbrace{p(x_0^k)}_{\mathcal{N}(x_0^k;0,I)}\;. \tag{15}$$

Substituting the transition (13) and dropping the constant $|\det H_\tau^k|$ gives the target used by the sampler,

$$\pi(x_1^k, x_0^k \mid y) \;\propto\; \underbrace{\exp\Big(-\tfrac{1}{2\eta^2}\|A_k x_1^k - y\|^2\Big)}_{p(y \mid x_1^k)\,:\ \text{likelihood}} \cdot \underbrace{p_\tau^k\big(\underbrace{H_\tau^k x_1^k + \sigma_\tau^k x_0^k}_{=\,x_\tau^k}\big)}_{p_\tau^k(x_\tau^k)\,:\ \text{flow marginal at time }\tau} \cdot \underbrace{\mathcal{N}(x_0^k; 0, I)}_{p(x_0^k)}. \tag{16}$$

Only p_τ^k is intractable in (16): it is represented by a network through its score and cannot be evaluated. The rest of the note is organised around isolating that factor.

Step 3: consistency. Marginalizing x_0^k out of (16) and applying Lemma 1(a) returns

$$\pi(x_1^k \mid y) \;\propto\; p(y \mid x_1^k)\, \widetilde{p}_\tau^k(x_1^k), \tag{17}$$

which is (10). The augmentation has therefore not changed what is being sampled: (16) is the joint whose x_1^k-marginal is the posterior we set out to target, and the blurring (9) remains the only approximation separating it from (8).

1.6 Plan

The rest of the note is a straight line from (16).

1. **Read off the two conditionals** of (16) by Bayes' rule (Section 1.7). Both of their gradients are computable from the network score.
2. **Apply a Langevin step to each.** With a preconditioner supplied in closed form by a curvature bound, this yields the sampler used in the paper (Section 1.8). Its one irreducible cost is a step size h_1 in the clean-image block, and with it a bias of order h_1.
3. **Remove h_1.** The relation $x_\tau^k = H_\tau^k x_1^k + \sigma_\tau^k x_0^k$ is linear, and conditioning on x_τ^k rather than x_0^k makes the clean-image conditional exactly Gaussian (Sections 2.1–2.2). It is then drawn exactly by one linear solve, with no step size and no bias.
4. **Evaluate the score** required by both routes. Tweedie's identity reduces it to $\mathbb{E}[X_0^k \mid x_\tau^k]$, which one network evaluation determines (Section 3).

Section 4 states both algorithms and compares what each leaves to be chosen.

1.7 The two conditionals

A Gibbs sampler for (16) needs its two conditionals, and Bayes' rule supplies both by inspection: hold the other variable fixed, keep every factor containing the variable of interest, and discard every factor that does not, since such a factor is a constant absorbed into the normalization.

Holding x_1^k fixed, the likelihood is such a constant:

$$\pi(x_0^k \mid x_1^k, y) \;\propto\; \underbrace{\cancel{\exp\Big(-\tfrac{1}{2\eta^2}\|A_k x_1^k - y\|^2\Big)}}_{\text{free of } x_0^k:\ \text{drops}} \cdot \underbrace{p_\tau^k\big(H_\tau^k x_1^k + \sigma_\tau^k x_0^k\big)}_{\text{contains } x_0^k} \cdot \underbrace{\mathcal{N}(x_0^k; 0, I)}_{\text{contains } x_0^k}. \tag{18}$$

This is the target of Block 2 in (32) below; the measurement does not enter, so Block 2 carries no data-consistency term.

Holding x_0^k fixed, nothing drops, and the intractable factor still moves with x_1^k:

$$\pi(x_1^k \mid x_0^k, y) \;\propto\; \underbrace{\exp\Big(-\tfrac{1}{2\eta^2}\|A_k x_1^k - y\|^2\Big)}_{\text{Gaussian in } x_1^k} \cdot \underbrace{p_\tau^k\big(H_\tau^k x_1^k + \sigma_\tau^k x_0^k\big)}_{\text{intractable; depends on } x_1^k}. \tag{19}$$

Both gradients are computable; that is not the issue. Differentiating (18) and (19) through $x_\tau^k = H_\tau^k x_1^k + \sigma_\tau^k x_0^k$ by the chain rule,

$$\nabla_{x_0^k} \log \pi(x_0^k \mid x_1^k, y) = \sigma_\tau^k\, \nabla \log p_\tau^k(x_\tau^k) \;-\; x_0^k, \tag{20}$$

$$\nabla_{x_1^k} \log \pi(x_1^k \mid x_0^k, y) = \tfrac{1}{\eta^2} A_k^\top\big(y - A_k x_1^k\big) \;+\; (H_\tau^k)^\top \nabla \log p_\tau^k(x_\tau^k), \tag{21}$$

and $\nabla \log p_\tau^k(x_\tau^k)$ is exactly what the network supplies. A Langevin step could therefore be applied to *both* variables in the coordinates (x_1^k, x_0^k), with no change of variables at all.

1.8 The sampler obtained directly: preconditioned Langevin

Equations (20)–(21) already define a sampler. Applying an unadjusted Langevin step to each conditional in turn gives

$$x_1^k \;\leftarrow\; x_1^k + P\Big[\tfrac{h_1}{2}\,\nabla_{x_1^k}\log\pi \;+\; \sqrt{h_1}\,\zeta\Big], \qquad \zeta\sim\mathcal{N}(0,P^{-1}), \tag{22}$$

$$x_0^k \;\leftarrow\; x_0^k + \tfrac{h_0}{2}\,\nabla_{x_0^k}\log\pi \;+\; \sqrt{h_0}\,\xi_0, \qquad \xi_0\sim\mathcal{N}(0,I_n), \tag{23}$$

with P a symmetric positive definite preconditioner. The noise in (22) has covariance $P\,(P^{-1})\,P = h_1P$ after the factor h_1, as preconditioned Langevin requires; writing it in this form, with ζ drawn from the *inverse* of the preconditioner and pushed through P, is what avoids ever forming $P^{1/2}$. Each step leaves its conditional invariant up to a discretization bias of order the step size, so the sweep is an unadjusted Langevin-within-Gibbs scheme for (16). This is the route taken in the paper.

The preconditioner is determined, not chosen. A preconditioner is admissible for any SPD P, but the useful choice dominates the curvature of the conditional. That curvature is bounded in closed form.

Proposition 2 (Curvature bound)**.** *For every x_1^k and x_0^k,*

$$\underbrace{\nabla^2_{x_1^k}\big(-\log\pi(x_1^k\mid x_0^k,y)\big)}_{\textit{curvature of the conditional}} \;\preceq\; \underbrace{\tfrac{1}{\eta^2}A_k^\top A_k}_{\textit{from } p(y\,\mid\,x_1^k)} \;+\underbrace{\tfrac{1}{(\sigma_\tau^k)^2}(H_\tau^k)^\top H_\tau^k}_{\textit{from } p_\tau^k} \;=:\; M_\tau. \tag{24}$$

The bound involves neither the data distribution, nor the network architecture, nor the quality of training, and no Lipschitz constant of the learned score is required. Taking

$$P := M_\tau^{-1} = \Big(\tfrac{1}{\eta^2}A_k^\top A_k + \tfrac{1}{(\sigma_\tau^k)^2}(H_\tau^k)^\top H_\tau^k\Big)^{-1} \tag{25}$$

therefore fixes the preconditioner from the stage schedule alone. Two consequences follow. The preconditioned curvature satisfies $M_\tau^{-1/2}\nabla^2(-\log\pi)M_\tau^{-1/2}\preceq I$, so (22) is stable for every $h_1\in(0,2)$, a certified range rather than one found by search. And M_τ is independent of the state, so it is fixed for all S inner iterations at a given (k,τ) and the noise in (22) may be generated without forming $M_\tau^{1/2}$, by the device of Lemma 5 below.

Why h_1 cannot be eliminated here. Within the coordinates (x_1^k, x_0^k) the step size is not a matter of tuning effort; it is forced by the form of (19). An exact draw from a density requires either conjugacy, or its normalizing constant. Here the factor $p_\tau^k(H_\tau^k x_1^k + \sigma_\tau^k x_0^k)$ is a learned density evaluated at a point that *moves with* x_1^k, so the conditional is not Gaussian, and only its score is available, never its value. Langevin is the appropriate tool in exactly that situation, and every Langevin scheme carries a step size. Adjusting the step by a Metropolis test would remove the bias but requires evaluating p_τ^k, which is unavailable; a proximal or semi-implicit treatment of the Gaussian factor removes the stability constraint but leaves a proximal parameter in its place.

The relation $x_\tau^k = H_\tau^k x_1^k + \sigma_\tau^k x_0^k$ is linear, and that linearity is what makes an exact draw possible — but it must be used, and using it means changing which variable is held fixed. Holding x_0^k fixed leaves the argument of p_τ^k moving with x_1^k; holding x_τ^k fixed freezes it. The two are related by a linear bijection, so no reformulation *within* (x_1^k, x_0^k) can achieve what the change to (x_1^k, x_τ^k) achieves: the obstruction is which variable is conditioned on, not the algebra. Section 2.2 carries this out.

The gradient in computable form. Equation (21) states the drift through $\nabla \log p_\tau^k$, which is not itself available. Substituting Prop. 6, $\nabla \log p_\tau^k(x_\tau^k) = -\widehat{x}_0^k/\sigma_\tau^k$, gives the form actually evaluated:

$$\nabla_{x_1^k} \log \pi(x_1^k \mid x_0^k, y) = \underbrace{\tfrac{1}{\eta^2} A_k^\top \big(y - A_k x_1^k\big)}_{\text{likelihood; depends on the current } x_1^k} - \underbrace{\tfrac{1}{\sigma_\tau^k} (H_\tau^k)^\top \, \widehat{x}_0^k}_{\text{prior; from one network evaluation}} . \tag{26}$$

This is line 13 of Algorithm 1. Both terms are matrix-free, and $\widehat{x}_0^k$ is the output of (38).

Cost: one solve, and no square root. With $P = M_\tau^{-1}$ the required noise is $\zeta \sim \mathcal{N}(0, M_\tau)$, and M_τ is a sum of two Gram matrices. A vector with exactly that covariance is therefore obtained by pushing white noise through the two adjoints,

$$\zeta \;=\; \underbrace{\tfrac{1}{\eta} A_k^\top \xi_y}_{\mathrm{Cov} = \frac{1}{\eta^2} A_k^\top A_k} \;+\; \underbrace{\tfrac{1}{\sigma_\tau^k} (H_\tau^k)^\top \xi_h}_{\mathrm{Cov} = \frac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top H_\tau^k}, \qquad \xi_y \sim \mathcal{N}(0, I_m), \quad \xi_h \sim \mathcal{N}(0, I_n) \text{ independent,} \tag{27}$$

so that $\mathrm{Cov}(\zeta) = M_\tau$. Substituting into (22), the drift and the noise share a single right-hand side and the update is one linear solve,

$$M_\tau \, \delta \;=\; \tfrac{h_1}{2} \nabla_{x_1^k} \log \pi \;+\; \sqrt{h_1}\, \zeta, \qquad x_1^k \leftarrow x_1^k + \delta, \tag{28}$$

solved by conjugate gradients using only the products $A_k v$, $A_k^\top w$, $H_\tau^k v$ and $(H_\tau^k)^\top v$. Neither M_τ, nor $P = M_\tau^{-1}$, nor $M_\tau^{1/2}$ is formed. The same device appears again in Lemma 5.

The noise-endpoint update (23) needs no such treatment. It is an unpreconditioned Langevin step: given $\widehat{x}_0^k$ it is an affine map of x_0^k plus a white-noise draw, with no operator to invert and no solve. Preconditioning it would gain little, since by (35) its curvature is I up to the contribution of p_τ^k, which the bound of Proposition 2 controls by 1 after the scaling by σ_τ^k.

The per-iteration cost is therefore one network evaluation and two linear solves, *neither of which belongs to* (23):

1. the clean-image update (28), which applies M_τ^{-1};
2. the score estimate $\widehat{x}_0^k$ of (38), which is an *input* to (23) and is used by (28) as well.

The exact draw of Section 2.2 has the same count, since it replaces the right-hand side of (28) rather than adding a solve.

The two solves are of different character. The operator in (28) mixes $A_k^\top A_k$ with $(H_\tau^k)^\top H_\tau^k$, which are in general diagonal in different bases, so it requires iteration. The operator in the score solve is a polynomial in G alone: it degenerates to the scalar $e_k - s_k$ when $G = I$, requiring no solve at all, and reduces to a pointwise division and two transforms whenever G is diagonalized by the discrete Fourier transform.

Even at $h_1 \to 2$, where the drift of (22) moves fully to the conditional mean of the Gaussian factor, the injected noise has covariance $2M_\tau^{-1}$ rather than M_τ^{-1}. The remaining discrepancy is in the noise scaling, and no choice of h_1 removes it.

2 An alternative route: an exact draw that removes h_1

Section 1.8 obtained a sampler by applying a Langevin step to each conditional, at the cost of a step size h_1 in the clean-image block and a bias of order h_1. This section removes both. A change of coordinates makes that conditional exactly Gaussian, after which it is drawn by a single linear solve. Nothing else about the target or about Block 2 changes, and by Table 3 the whole difference in what must be chosen is the one quantity h_1.

2.1 The change of variables

Block 1 will need (16) written in the pair (x_1^k, x_τ^k) rather than (x_1^k, x_0^k). Substituting $x_0^k = (x_\tau^k - H_\tau^k x_1^k)/\sigma_\tau^k$ from (2), with $dx_0^k = (\sigma_\tau^k)^{-n} dx_\tau^k$ at fixed x_1^k, the third factor becomes

$$\underbrace{\mathcal{N}(x_0^k; 0, I)}_{\text{density in } x_0^k} \cdot (\sigma_\tau^k)^{-n} = \mathcal{N}\big(x_\tau^k - H_\tau^k x_1^k; 0, (\sigma_\tau^k)^2 I\big) = \mathcal{N}\big(x_\tau^k; H_\tau^k x_1^k, (\sigma_\tau^k)^2 I\big), \tag{29}$$

the second factor becomes $p_\tau^k(x_\tau^k)$, and the first is unchanged. Hence

$$\pi(x_1^k, x_\tau^k \mid y) \;\propto\; \underbrace{p(y \mid x_1^k)}_{\text{Gaussian in } x_1^k} \cdot \underbrace{\mathcal{N}\big(x_\tau^k; H_\tau^k x_1^k, (\sigma_\tau^k)^2 I\big)}_{\text{Gaussian in } x_1^k} \cdot \underbrace{p_\tau^k(x_\tau^k)}_{\text{free of } x_1^k}. \tag{30}$$

In (30) the intractable factor p_τ^k depends on x_τ^k alone, so conditioning on x_τ^k removes it and leaves a product of two Gaussians in x_1^k. This is exactly the obstruction of (19), now removed. The substitution is also invertible, $x_0^k = (x_\tau^k - H_\tau^k x_1^k)/\sigma_\tau^k$, and that inverse is the line used in Algorithms 1 and 2 to restore the stored pair after x_1^k changes.

Conditioning a joint density on one of its arguments is Bayes' rule, and the two conditionals are all a Gibbs sampler needs. Each conditional draw leaves π invariant, so the alternation does; when a conditional cannot be drawn directly it is replaced by a Markov kernel leaving it invariant, such as a Langevin step. At each grid point (k, τ) and for each of S inner iterations the sampler performs

$$\text{Block 1:} \quad x_1^k \;\leftarrow\; \text{exact draw from } \pi(x_1^k \mid x_\tau^k, y), \qquad x_\tau^k \text{ held fixed}, \tag{31}$$

$$\text{Block 2:} \quad x_0^k \;\leftarrow\; \text{one Langevin step for } \pi(x_0^k \mid x_1^k, y), \qquad x_1^k \text{ held fixed}. \tag{32}$$

Because the state is stored as (x_1^k, x_0^k), Block 1 is followed by $x_0^k \leftarrow (x_\tau^k - H_\tau^k x_1^k)/\sigma_\tau^k$, which is the inverse of the substitution in (29) and keeps x_τ^k unchanged.

The two blocks read the same joint density in different coordinates: Block 2 uses (18) directly, and Block 1 uses (30), the same density in the coordinates (x_1^k, x_τ^k). Both are conditionals of π, so each leaves it invariant and the alternation is a valid sweep. The measurement is kept in the conditioning of (32) only for symmetry with (31); by (18) it drops out, so $\pi(x_0^k \mid x_1^k, y) = \pi(x_0^k \mid x_1^k)$.

2.2 The exact draw

Proposition 3 (The clean-image conditional is Gaussian)**.** *Let*

$$\underbrace{M_\tau := \tfrac{1}{\eta^2} A_k^\top A_k + \tfrac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top H_\tau^k}_{\textit{precision; independent of the state}}, \qquad \underbrace{b_\tau := \tfrac{1}{\eta^2} A_k^\top y + \tfrac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top x_\tau^k}_{\textit{information vector; depends on } y,\ x_\tau^k}. \tag{33}$$

If M_τ is nonsingular, then $x_1^k \mid x_\tau^k, y \sim \mathcal{N}\big(M_\tau^{-1} b_\tau,\ M_\tau^{-1}\big)$.

Note that both terms of M_τ are dictated by (16); neither is a chosen weight. M_τ is symmetric positive semidefinite, being a sum of two Gram matrices, and it fails to be positive definite exactly in the situation described next.

Proposition 4 (A ridge is required at $\tau = 0$)**.** *Let $\mathcal{K}_\tau := \ker(H_\tau^k) \cap \ker(A_k)$. The target (16) is constant along $\mathcal{K}_\tau$ and is therefore not normalizable whenever $\mathcal{K}_\tau \neq \{0\}$. At the start of every stage, $\tau = 0$ and $H_0^k = s_k G$, which is rank-deficient.*

A ridge ϵI with $\epsilon > 0$ is therefore added to M_τ. It is required by the formulation, not by numerics.

Drawing from $\mathcal{N}(M_\tau^{-1} b_\tau, M_\tau^{-1})$ appears to need $M_\tau^{-1/2}$, which is unavailable for the operators of interest. It is not needed. The key observation is that M_τ is a sum of two Gram matrices, so a vector with covariance exactly M_τ is obtained by pushing white noise through the two adjoints and adding; solving against M_τ then turns covariance M_τ into covariance M_τ^{-1}.

Lemma 5 (Exact sampling by a single linear solve)**.** *Let $\xi_y \sim \mathcal{N}(0, I_m)$ and $\xi_h \sim \mathcal{N}(0, I_n)$ be independent, and set*

$$\widetilde{b}_\tau := b_\tau + \underbrace{\frac{1}{\eta} A_k^\top \xi_y + \frac{1}{\sigma_\tau^k} (H_\tau^k)^\top \xi_h}_{\zeta,\ \mathrm{Cov}(\zeta) = M_\tau} . \tag{34}$$

Then $x_1^k := M_\tau^{-1} \widetilde{b}_\tau$ has the law $\mathcal{N}\big(M_\tau^{-1} b_\tau, M_\tau^{-1}\big)$ of Proposition 3.

The draw is exact for every τ, so Block 1 has no step size. The system $M_\tau x_1^k = \widetilde{b}_\tau$ is symmetric positive definite and is solved by conjugate gradients, which needs only the products $A_k v$, $A_k^\top w$, $H_\tau^k v$ and $(H_\tau^k)^\top v$; neither M_τ nor its inverse nor its square root is formed.

3 The score for the noise endpoint

By (30) the conditional for x_0^k retains the factor p_τ^k and is not Gaussian, so a Langevin step is used. Differentiating $\log \pi$ of (16) in x_0^k,

$$\nabla_{x_0^k} \log \pi = \underbrace{\sigma_\tau^k \, \nabla_{x_\tau^k} \log p_\tau^k(x_\tau^k)}_{\text{score of the flow marginal}} - \underbrace{x_0^k}_{\text{Gaussian factor}} . \tag{35}$$

It remains to evaluate the score. Proposition 6 reduces it to a conditional mean, and Proposition 7 computes that mean from one network evaluation.

Proposition 6 (Score in noise form)**.** $\nabla_{x_\tau^k} \log p_\tau^k(x_\tau^k) = -\,\mathbb{E}[X_0^k \mid x_\tau^k]/\sigma_\tau^k$.

Writing $\widehat{x}_0^k$ for the estimate of $\mathbb{E}[X_0^k \mid x_\tau^k]$, the Block-2 update is therefore

$$x_0^k \;\leftarrow\; x_0^k - \frac{h_0}{2}\big(x_0^k + \widehat{x}_0^k\big) + \sqrt{h_0}\, \xi_0, \qquad \xi_0 \sim \mathcal{N}(0, I_n). \tag{36}$$

Expressing the score through X_0^k rather than X_1^k divides by σ_τ^k once rather than twice and avoids differencing two vectors that become close as $\sigma_\tau^k \to 0$, so the cancellation at the end of the final stage is removed.

Proposition 7 (The conditional mean of the noise endpoint)**.** *Write $\bar{x}_1^k := \mathbb{E}[X_1^k \mid x_\tau^k]$ and $\bar{x}_0^k := \mathbb{E}[X_0^k \mid x_\tau^k]$. Then*

$$\boxed{\underbrace{\big[e_k(1 - s_k) I - s_k(1 - e_k) G\big]}_{N_k} \, \bar{x}_0^k \;=\; \underbrace{B_k \, x_\tau^k - H_\tau^k \, d_\tau^k}_{\textit{computable from } x_\tau^k \textit{ and } v_\theta} .} \tag{37}$$

Like B_k, the operator N_k is a polynomial in G and does not depend on τ, so it is constant throughout a stage. Equation (37) is exact when the network returns d_τ^k exactly; Corollary 8 accounts for the network error and gives the system solved in Algorithm 2.

Corollary 8 (Regularized estimator)**.** *Suppose $v_\theta(x_\tau^k, \tau, k) = d_\tau^k + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \gamma^2 I)$ independent of X_0^k, and take $X_0^k \sim \mathcal{N}(0, I)$. Write $z_\tau^k := B_k x_\tau^k - H_\tau^k v_\theta$. Then $z_\tau^k \mid X_0^k \sim \mathcal{N}\big(N_k X_0^k,\ \gamma^2 (H_\tau^k)^2\big)$, and the posterior mean of X_0^k given z_τ^k satisfies*

$$\boxed{\Big[N_k^2 + \gamma^2 (H_\tau^k)^2\Big]\widehat{x}_0^k \;=\; N_k\, z_\tau^k \;=\; N_k\Big(B_k\, x_\tau^k - H_\tau^k\, v_\theta(x_\tau^k, \tau, k)\Big).} \tag{38}$$

The prior used here is the marginal law $\mathcal{N}(0, I)$ of X_0^k rather than its conditional law given x_τ^k, so (38) is a shrinkage approximation; its defining assumption nonetheless holds by construction, and the ridge $\gamma^2 (H_\tau^k)^2$ bounds the condition number of the system.

The right-hand side in terms of the endpoint reconstructions. The quantity $z_\tau^k = B_k x_\tau^k - H_\tau^k v_\theta$ appearing on the right of (38) has a simpler form. Substituting $v_\theta = \widehat{x}_{\text{end}}^k - \widehat{x}_{\text{start}}^k$ and $x_\tau^k = (1-\tau)\widehat{x}_{\text{start}}^k + \tau \widehat{x}_{\text{end}}^k$ from (6), the coefficients collapse, $(1-\tau)B_k + H_\tau^k = e_k I$ and $\tau B_k - H_\tau^k = -s_k G$, so

$$z_\tau^k \;=\; B_k\, x_\tau^k - H_\tau^k\, v_\theta \;=\; e_k\, \widehat{x}_{\text{start}}^k - s_k\, G\, \widehat{x}_{\text{end}}^k. \tag{39}$$

The two eliminations are then visibly symmetric, each a cross-combination of the same two reconstructions:

$$\underbrace{N_k\, \bar{x}_0^k = e_k \widehat{x}_{\text{start}}^k - s_k G\, \widehat{x}_{\text{end}}^k,}_{\text{noise endpoint, (37)}} \qquad \underbrace{M_\tau^{\text{ep}}\, \bar{x}_1^k = (1-e_k)\widehat{x}_{\text{start}}^k - (1-s_k)\widehat{x}_{\text{end}}^k,}_{\text{clean endpoint, (46) below}} \tag{40}$$

with $N_k = e_k(1-s_k)I - s_k(1-e_k)G$ and $M_\tau^{\text{ep}} = (1-e_k)s_k G - (1-s_k)e_k I$. Both operators reduce to scalars when $G = I$: the first to $e_k - s_k$, which turns (39) into $\widehat{x}_0^k = (e_k \widehat{x}_{\text{start}}^k - s_k \widehat{x}_{\text{end}}^k)/(e_k - s_k)$, which is Equation (7) of the submission. Writing the right-hand side as (39) rather than $B_k x_\tau^k - H_\tau^k v_\theta$ therefore makes the comparison with that equation immediate: the only difference is the G multiplying $\widehat{x}_{\text{end}}^k$, and the operator on the left.

Why (38) is more accurate than the estimators it replaces. Two other endpoint estimators appear in this setting, and (38) replaces both. The first is a weighted least-squares fit, (45) below, which treats $\widehat{x}_{\text{start}}^k$ and $\widehat{x}_{\text{end}}^k$ as independently corrupted observations of x_1^k and does not solve for x_0^k at all. The second is a closed form obtained by inverting the boundary relations as if G were the identity. Section 6.1 examines both quantitatively; the reason (38) is preferable can be stated here.

The two relations of (1) share the *same* noise endpoint x_0^k. That is what makes the pair determined: two vector equations in the two unknowns (x_1^k, x_0^k). Eliminating one of them algebraically therefore cancels the noise *exactly*, which is what Prop. 7 does and what (38) regularizes. A least-squares fit that treats the two residuals as independent noise to be averaged down discards that structure, and is then left with nothing to average: the residuals are collinear, since $(1-e_k)\big(s_k G x_1^k - \widehat{x}_{\text{start}}^k\big) = (1-s_k)\big(e_k x_1^k - \widehat{x}_{\text{end}}^k\big)$ identically. The closed form retains the structure but discards G, which mis-scales the component in $\ker(G)$.

Table 2: The three endpoint estimators. Only (38) is exact in the limit of an exact network while also retaining the blur, and only its regularizer rests on an assumption that holds by construction.

Estimator	Exact when the network is exact?	Retains G?	Regularizer
weighted least squares, (45)	**no**: biased by $\beta_k x_0^k$, with $\beta_k = 3$ at the coarsest stage	yes	λ_{wls}, tuned; imposes a Gaussian prior on images
closed form, Eq. (7) of the submission	only if $G = I$	**no**	none
(38), used here	**yes**, as $\gamma \to 0$ (Prop. 7)	yes	$\gamma^2 (H_\tau^k)^2$, from a measurable variance

The last column matters as much as the first. The ridge in (38) is $\gamma^2(H_\tau^k)^2$, where γ^2 is the variance of the velocity error and can be measured on held-out data by (56); its defining assumption, $X_0^k \sim \mathcal{N}(0, I)$, holds by construction. The ridge λ_{wls} instead imposes a Gaussian prior on natural images, which is not believed and therefore has to be tuned.

4 The two algorithms

Algorithm 1 is the scheme of Section 1.8, with the preconditioner (25) supplied by Proposition 2. Algorithm 2 replaces its Block 1 by the exact draw of Section 2.2. They differ in three lines and share everything else, including Block 2 and the score evaluation.

Algorithm 1 Preconditioned Langevin within Gibbs. Both blocks are Langevin steps; h_1 is required. The red box is the score solve, (38); the blue box is the clean-image solve. Their operators differ: the first is a polynomial in G alone, the second mixes $A_k^\top A_k$ with $(H_\tau^k)^2$.

Require: y, A, η, stages K, time grid $\{\tau\}$, inner iterations S, step sizes $h_1 \in (0, 2)$ and h_0, velocity-error variance γ^2, CG iterations L, ridge $\epsilon > 0$

$x_1^0 \leftarrow 0$
for $k = 0, \ldots, K-1$ **do**
  $x_0^k \sim \mathcal{N}(0, I_n)$
  for τ on the stage grid **do**
    $H_\tau^k \leftarrow (1-\tau)s_k G + \tau e_k I;\quad \sigma_\tau^k \leftarrow (1-\tau)(1-s_k) + \tau(1-e_k)$
    $B_k \leftarrow e_k I - s_k G;\quad N_k \leftarrow e_k(1-s_k)I - s_k(1-e_k)G$
    $M_\tau \leftarrow \frac{1}{\eta^2} A_k^\top A_k + \frac{1}{(\sigma_\tau^k)^2}(H_\tau^k)^\top H_\tau^k + \epsilon I$ ▷ (25); no λ_{pre}
    for $s = 1, \ldots, S$ **do**
      $x_\tau^k \leftarrow H_\tau^k x_1^k + \sigma_\tau^k x_0^k$
  Score
      $v \leftarrow v_\theta(x_\tau^k, \tau, k)$ ▷ one network evaluation
      $\boxed{\text{solve } \left[N_k^2 + \gamma^2 (H_\tau^k)^2\right] \widehat{x}_0^k = N_k\left(B_k x_\tau^k - H_\tau^k v\right)}$ ▷ **score solve**, Cor. 8
  Block 1: preconditioned Langevin step for x_1^k
      $\xi_y \sim \mathcal{N}(0, I_m);\quad \xi_h \sim \mathcal{N}(0, I_n)$
      $g \leftarrow \frac{1}{\eta^2} A_k^\top (y - A_k x_1^k) - \frac{1}{\sigma_\tau^k}(H_\tau^k)^\top \widehat{x}_0^k$ ▷ (21), Prop. 6
      $\boxed{\text{solve } M_\tau\, \delta = \frac{h_1}{2}\, g + \sqrt{h_1}\left(\frac{1}{\eta} A_k^\top \xi_y + \frac{1}{\sigma_\tau^k}(H_\tau^k)^\top \xi_h\right)}$ ▷ **clean-image solve**, (28)
      $x_1^k \leftarrow x_1^k + \delta$ ▷ invariance up to $O(h_1)$
  Block 2: Langevin step for x_0^k *(no solve)*
      $\xi_0 \sim \mathcal{N}(0, I_n)$
      $x_0^k \leftarrow x_0^k - \frac{h_0}{2}\left(x_0^k + \widehat{x}_0^k\right) + \sqrt{h_0}\, \xi_0$ ▷ (36)
    end for
end inner iterations s
  end for
end time grid τ (stage k complete)
  $x_1^{k+1} \leftarrow U^{(1)}(x_1^k);\quad x_0^{k+1} \sim \mathcal{N}(0, I)$ ▷ ancestral transition to the next scale
end for
end scales k
return x_1^{K-1}

Algorithm 2 Exact conditional draw for x_1^k, Langevin step for x_0^k; no h_1. The red and blue boxes are the same two solves as in Algorithm 1: the red one is identical, and the blue one uses the same operator M_τ with a different right-hand side, (42) rather than (41).

Require: y, A, η, stages K, time grid $\{\tau\}$, inner iterations S, step size h_0, velocity-error variance γ^2, CG iterations L, ridge $\epsilon > 0$

$x_1^0 \leftarrow 0$
for $k = 0, \ldots, K-1$ **do**
  $x_0^k \sim \mathcal{N}(0, I_n)$
  for τ on the stage grid **do**
    $H_\tau^k \leftarrow (1-\tau)s_k G + \tau e_k I; \quad \sigma_\tau^k \leftarrow (1-\tau)(1-s_k) + \tau(1-e_k)$
    $B_k \leftarrow e_k I - s_k G; \quad N_k \leftarrow e_k(1-s_k)I - s_k(1-e_k)G$
    $M_\tau \leftarrow \frac{1}{\eta^2} A_k^\top A_k + \frac{1}{(\sigma_\tau^k)^2}(H_\tau^k)^\top H_\tau^k + \epsilon I$ ▷ (33); ϵ by Prop. 4
    for $s = 1, \ldots, S$ **do**
      $x_\tau^k \leftarrow H_\tau^k x_1^k + \sigma_\tau^k x_0^k$
  Score — identical to Algorithm 1
      $v \leftarrow v_\theta(x_\tau^k, \tau, k)$ ▷ one network evaluation
      $\boxed{\text{solve } \big[N_k^2 + \gamma^2 (H_\tau^k)^2\big]\widehat{x}_0^k = N_k\big(B_k x_\tau^k - H_\tau^k v\big)}$ ▷ **score solve**, Cor. 8
  Block 1: exact draw of x_1^k from $\pi(x_1^k \mid x_\tau^k, y)$
      $\xi_y \sim \mathcal{N}(0, I_m); \quad \xi_h \sim \mathcal{N}(0, I_n)$
      $\widetilde{b}_\tau \leftarrow \frac{1}{\eta^2} A_k^\top y + \frac{1}{(\sigma_\tau^k)^2}(H_\tau^k)^\top x_\tau^k + \frac{1}{\eta} A_k^\top \xi_y + \frac{1}{\sigma_\tau^k}(H_\tau^k)^\top \xi_h$ ▷ (34)
      $\boxed{\text{solve } M_\tau\, x_1^k = \widetilde{b}_\tau}$ ▷ **clean-image solve**, Lem. 5; exact, no h_1
      $x_0^k \leftarrow (x_\tau^k - H_\tau^k x_1^k)/\sigma_\tau^k$ ▷ inverse of (29); x_τ^k unchanged
  Block 2: Langevin step for x_0^k *(no solve)* — identical
      $\xi_0 \sim \mathcal{N}(0, I_n)$
      $x_0^k \leftarrow x_0^k - \frac{h_0}{2}\big(x_0^k + \widehat{x}_0^k\big) + \sqrt{h_0}\,\xi_0$ ▷ (36)
    end for
end inner iterations s
  end for
end time grid τ (stage k complete)
  $x_1^{k+1} \leftarrow U^{(1)}(x_1^k); \quad x_0^{k+1} \sim \mathcal{N}(0, I)$ ▷ ancestral transition to the next scale
end for
end scales k
return x_1^{K-1}

The two Block-1 updates, and why their prior terms have opposite signs. Both solve against the same operator M_τ, but their right-hand sides are the gradients of *different* conditionals. Writing each as an increment $x_1^k \leftarrow x_1^k + \delta$,

$$\text{Alg. 1:} \quad M_\tau\,\delta = \tfrac{h_1}{2}\,\nabla_{x_1^k} \log \pi\big(x_1^k \mid x_0^k, y\big) \; + \; \sqrt{h_1}\,\zeta, \tag{41}$$

$$\text{Alg. 2:} \quad M_\tau\,\delta = \nabla_{x_1^k} \log \pi\big(x_1^k \mid x_\tau^k, y\big) \; + \; \zeta, \tag{42}$$

the second being $M_\tau x_1^k = \widetilde{b}_\tau$ rewritten, since $\nabla_{x_1^k} \log \pi(x_1^k \mid x_\tau^k, y) = b_\tau - M_\tau x_1^k$ by (62). Expanding the two gradients,

$$\nabla_{x_1^k} \log \pi\big(x_1^k \mid x_0^k, y\big) = \tfrac{1}{\eta^2} A_k^\top (y - A_k x_1^k) \underbrace{- \tfrac{1}{\sigma_\tau^k} (H_\tau^k)^\top \widehat{x}_0^k}_{\text{estimated}}, \tag{43}$$

$$\nabla_{x_1^k} \log \pi\big(x_1^k \mid x_\tau^k, y\big) = \tfrac{1}{\eta^2} A_k^\top (y - A_k x_1^k) \underbrace{+ \tfrac{1}{\sigma_\tau^k} (H_\tau^k)^\top x_0^k}_{\text{current state}}. \tag{44}$$

The opposite signs are not an inconsistency. The two conditionals hold different variables fixed, so they are different functions of x_1^k, and their gradients need not agree — and do not, even when the estimate is perfect, $\widehat{x}_0^k = x_0^k$. Concretely:

In $\pi(x_1^k \mid x_\tau^k, y)$ the interpolant is fixed, so changing x_1^k changes the residual $x_\tau^k - H_\tau^k x_1^k = \sigma_\tau^k x_0^k$. The prior contribution is the coupling $\mathcal{N}(x_\tau^k; H_\tau^k x_1^k, (\sigma_\tau^k)^2 I)$, whose gradient $\frac{1}{(\sigma_\tau^k)^2}(H_\tau^k)^\top (x_\tau^k - H_\tau^k x_1^k) = +\frac{1}{\sigma_\tau^k}(H_\tau^k)^\top x_0^k$ pulls x_1^k toward $(H_\tau^k)^{-1} x_\tau^k$. Here x_0^k is a residual to be *reduced*, so it enters with a positive sign.

In $\pi(x_1^k \mid x_0^k, y)$ the noise endpoint is fixed and the interpolant moves with x_1^k. The prior contribution is $\log p_\tau^k(H_\tau^k x_1^k + \sigma_\tau^k x_0^k)$, whose gradient $(H_\tau^k)^\top \nabla \log p_\tau^k = -\frac{1}{\sigma_\tau^k}(H_\tau^k)^\top \widehat{x}_0^k$ pushes x_1^k in the direction that makes x_τ^k more probable under the flow, that is toward the denoised $H_\tau^k \widehat{x}_1^k$. Here $\widehat{x}_0^k$ is the noise the network reports as *present* in x_τ^k, and the drift removes it, so it enters with a negative sign.

Two further remarks. Conjugate gradients is linear, so the scalar $h_1/2$ in (41) may be moved outside the solve; this does not reconcile the two updates, whose right-hand sides differ as vectors and not merely by a scale. And M_τ is not the operator of the other solve: each algorithm performs two solves per iteration with *different* operators, $N_k^2 + \gamma^2 (H_\tau^k)^2$ for the score (38), a polynomial in G alone, and M_τ for Block 1, which mixes $A_k^\top A_k$ with $(H_\tau^k)^2$. Across the two algorithms both operators are identical; within one algorithm they are not.

Finally, M_τ plays a different role in each. In (41) it is a *preconditioner*, a metric selected by Prop. 2; in (42) it is the *precision* of the Gaussian conditional of Prop. 3. That the two coincide is the content of the curvature bound, and it is why the algorithms cost the same. The drift also carries $h_1/2$ in one and not the other, and the noise $\sqrt{h_1}$ rather than 1; at $h_1 = 2$ the drift coefficients agree but the noise does not, which is the $2M_\tau^{-1}$ inflation noted in Section 1.8.

Block 1 leaves $\pi(\cdot \mid x_\tau^k, y)$ invariant exactly, by Proposition 3 and Lemma 5. Block 2 leaves $\pi(\cdot \mid x_1^k, y)$ invariant up to the discretization bias of the unadjusted Langevin algorithm, of order h_0, and up to the error in $\widehat{x}_0^k$. The sweep is a Metropolis-within-Gibbs scheme for (16) in which one of the two kernels is exact.

Table 3 contrasts the two. The only quantity present in one and absent in the other is h_1; everything else, including the operator M_τ, is shared.

Table 3 of Section 5 compares what each scheme leaves to be chosen.

In Algorithm 2 two continuous quantities are chosen: the Langevin step size h_0, and the velocity-error variance γ^2, which is an estimate of a measurable error rather than a free parameter. The ridge ϵ is required only at $\tau = 0$. Every other weight in Algorithm 2, in particular the two terms of M_τ, is dictated by (16) and the stage schedule.

Two assumptions should be checked against the implementation. The operator G must be symmetric, since Propositions 7 and Corollary 8 use commutativity of polynomials in G. And the inter-stage transition, upsampling x_1^k and drawing $x_0^{k+1} \sim \mathcal{N}(0, I)$ independently, should match the construction used to train the cascaded prior; if it does, that transition is the exact ancestral step.

5 How this addresses the points raised in review

Table 3: What each scheme leaves to be chosen. "NeurIPS version" is the sampler as submitted; values are those reported in its appendix. "Derived" means computed from the stage schedule (s_k, e_k, τ) with no free parameter. "Set by compute budget" means the quantity is raised until performance saturates or the budget is exhausted, rather than searched for an optimum; "set by CG tolerance" means L is chosen so the linear solve meets a fixed residual tolerance. By (53) the ridge ϵ is required only at $\tau = 0$, where $H_0^k = s_k G$ is singular; M_τ is positive definite for every $\tau > 0$. The counts in the last rows exclude γ^2: it is not a free quantity but the variance of the velocity error at scale k and time τ, obtained from (56) and tabulated with the model. Section 7.12 states the condition under which treating it as measured is legitimate, and the test that verifies it.

Quantity	NeurIPS version	Alg. 1 (Langevin)	Alg. 2 (exact draw)
Step sizes			
h_1, Block-1 step	tuned per task, 0.1–0.3	tuned, certified range $(0, 2)$	**not present**
h_0, Block-2 step	tuned per task, 10^{-3}–10^{-2}	tuned	tuned
Weights in the Block-1 update			
Preconditioner	$\lambda_{\mathrm{pre}} I$, tuned, 150–350	$(H_\tau^k)^\top H_\tau^k / (\sigma_\tau^k)^2$, derived	$(H_\tau^k)^\top H_\tau^k / (\sigma_\tau^k)^2$, derived
Anchor	λ_{anc}, tuned, 150–350	**not present**: duplicates the score term	**not present**
ϵ, ridge	absorbed into λ_{pre}	**only at** $\tau = 0$; unnecessary if the grid starts at $\tau > 0$	**only at** $\tau = 0$; unnecessary if the grid starts at $\tau > 0$
Endpoint estimate feeding the score			
Estimator	weighted least squares for $\widehat{x}_1^k$	(38) for $\widehat{x}_0^k$	(38) for $\widehat{x}_0^k$
Its regularizer	λ_{wls}, tuned	$\gamma^2 (H_\tau^k)^2$	$\gamma^2 (H_\tau^k)^2$
γ^2, velocity-error variance	**not present**: the ridge is λ_{wls}, a tuned scalar	determined by the scale-dependent noise variance $\gamma^2(k, \tau)$; **measured** by (56), not searched	determined by $\gamma^2(k, \tau)$; measured, not searched
Weights ρ_s^k, ρ_e^k	fixed to 1, unmodelled	**not present**	**not present**
Budget			
S, inner iterations	tuned, 10–15	set by compute budget	set by compute budget
L, CG iterations	set by CG tolerance	set by CG tolerance	set by CG tolerance
Totals and guarantees			
Continuous quantities tuned	**5**	**2**	**1**
Task-specific configurations	2 (perceptual, PSNR)	1	1
Block-1 update targets (16)	no: the anchor tilts the conditional	yes, up to $O(h_1)$	yes, exactly
Linear solves per iteration	2	2	2

Table 4 maps the points raised on the submitted version to the results that address them. Two of the entries are settled by the derivation alone and apply to Algorithm 1 as well, so they do not require adopting the exact draw.

Table 4: Points raised in review, and where each is addressed.

Point raised	Addressed by
The method is not principled as claimed; there is a gap between the algorithm and the posterior sampling theory	Sections 1.3–1.5. The sampler targets (10), an exact posterior. Exactly one approximation is made, the blurring (9), and it is stated before anything is derived
What distribution is sampled at the final stage?	(10) with $\widetilde{p}_\tau^k$ of (9). At the terminal step $\sigma_\tau^k = 0$ and $H_\tau^k = I$, so $\widetilde{p}_\tau^k = p_1^k$ and the target is (8)
The benefit is convoluted with hyperparameter tuning	Table 3: five tuned continuous quantities reduced to two, and to one under Algorithm 2; two task-specific configurations reduced to one
The updates are not Gibbs updates; a Gibbs sampler would draw from conditionals	Section 1.7 derives both conditionals; Algorithm 2 draws one of them exactly (Prop. 3, Lem. 5)
Heuristic components: anchor term, preconditioner weight, endpoint regularizer	The anchor duplicates a term already in the score and is removed (Section 1.8); λ_{pre} is replaced by $(H_\tau^k)^\top H_\tau^k/(\sigma_\tau^k)^2$ from Prop. 2; the endpoint ridge becomes $\gamma^2(H_\tau^k)^2$ from the velocity-error variance (Cor. 8)
The plug-in endpoint estimate is not analysed	Prop. 7 is exact given an exact network; Cor. 8 isolates the network error in the single quantity γ^2, which is measured rather than tuned (Section 7)
Approximations leave the true posterior intractable	The approximations are enumerated: the blur (9), the plug-in γ^2, and, in Algorithm 1 only, the step size h_1. Algorithm 2 removes the third
Computational cost is not reported	One network evaluation and two linear solves per inner iteration, identical for both algorithms (Section 1.8)

6 What may have gone wrong in the submitted implementation

This section collects four specific hypotheses about why the submitted sampler underperformed. They are derived from the equations and the hyperparameter values reported in that version, not from its code, so each is stated together with the measurement that would confirm or refute it. They are ordered by how much of the observed behaviour they could explain.

6.1 The endpoint estimator

The submission contains *two* different endpoint estimators, and neither is the one this note uses. We state all three, since the differences between them are the likely source of a large part of the error.

(i) The weighted least-squares estimator. The submission models $\widehat{x}_{\text{start}}^k$ and $\widehat{x}_{\text{end}}^k$ as two independently corrupted linear observations of x_1^k, with variances proportional to $(1-s_k)^2/\rho_s^k$ and $(1-e_k)^2/\rho_e^k$, places a ridge λ_{wls} on x_1^k, and minimizes

$$\widehat{x}_1^{\text{wls}} = \arg\min_x \frac{\rho_s^k}{(1-s_k)^2}\left\|s_k G x - \widehat{x}_{\text{start}}^k\right\|^2 + \frac{\rho_e^k}{(1-e_k)^2}\left\|e_k x - \widehat{x}_{\text{end}}^k\right\|^2 + \lambda_{\text{wls}}\|x\|^2. \quad (45)$$

The noise endpoint x_0^k appears nowhere: it is treated as nuisance noise to be averaged away.

(ii) Equation (7). A closed form, $\widehat{x}_1 = \big[(1-s_k)\widehat{x}^k_{\text{end}} - (1-e_k)\widehat{x}^k_{\text{start}}\big]/(e_k - s_k)$, with a companion expression for $\widehat{x}_0^k$.

(iii) The estimator used here. Proposition 7 and Corollary 8: eliminate $\bar{x}_1^k$ between the two conditional-mean relations and solve for $\bar{x}_0^k$, using the fact that the *same* x_0^k appears in both relations of (1).

Equation (7) is (iii) with the blur discarded. Eliminating x_0^k instead of x_1^k from (1) gives the companion of (37),

$$\underbrace{\big[(1-e_k)s_kG - (1-s_k)e_kI\big]}_{=:\ M_\tau^{\text{ep}}}\,\widehat{x}_1^k = (1-e_k)\widehat{x}^k_{\text{start}} - (1-s_k)\widehat{x}^k_{\text{end}}. \tag{46}$$

Setting $G = I$ collapses the operator to a scalar, $M_\tau^{\text{ep}} = (1-e_k)s_k - (1-s_k)e_k = s_k - e_k$, and dividing by it reproduces Equation (7) exactly. So Equation (7) is the correct elimination with the blur dropped, and its only error is that M_τ^{ep} is replaced by its value on the eigenspace of G with eigenvalue one. On $\ker(G)$ the correct eigenvalue is $-(1-s_k)e_k$, so the $\ker(G)$ component is scaled by $(1-s_k)e_k/(e_k-s_k)$, which for four evenly spaced stages is 1.0, 1.5, 1.5, 1.0. The error is systematic, so it does not average out, and it lives in the high-frequency band the cascade exists to reconstruct.

The weighted least-squares estimator is biased even when $G = I$. This is the more serious problem, and it does not require the blur to appear at all. Supply (45) with *exact* endpoints, $\widehat{x}^k_{\text{start}} = s_kx_1^k + (1-s_k)x_0^k$ and $\widehat{x}^k_{\text{end}} = e_kx_1^k + (1-e_k)x_0^k$ at $G = I$, and take $\lambda_{\text{wls}} = 0$, $\rho_s^k = \rho_e^k = 1$. The minimizer of (45) is then

$$\widehat{x}_1^{\text{wls}} = x_1^k + \beta_k\, x_0^k, \qquad \beta_k = \frac{\dfrac{s_k}{1-s_k} + \dfrac{e_k}{1-e_k}}{\dfrac{s_k^2}{(1-s_k)^2} + \dfrac{e_k^2}{(1-e_k)^2}}, \tag{47}$$

and $\beta_k > 0$ for every admissible (s_k, e_k) with $e_k < 1$. For four evenly spaced stages:

stage k (s_k, e_k)	0 $(0, \frac{1}{4})$	1 $(\frac{1}{4}, \frac{1}{2})$	2 $(\frac{1}{2}, \frac{3}{4})$	3 $(\frac{3}{4}, 1)$
bias β_k of (45)	**3.00**	**1.20**	0.40	0
error of Equation (7)	0	0	0	0

At the coarsest stage the weighted least-squares estimate of the clean endpoint is wrong by three times the noise endpoint, with exact inputs and no blur. Equation (7), by contrast, is exact under the same conditions.

The reason is structural. The two relations of (1) share the *same* x_0^k, so x_0^k can be eliminated algebraically and the noise cancels exactly; that is what (ii) and (iii) do. Equation (45) instead treats the two residuals as independent noise to be averaged down, and with only two observations there is nothing to average: the residuals are collinear, since $(1-e_k)\big(s_kGx_1^k - \widehat{x}^k_{\text{start}}\big) = (1-s_k)\big(e_kx_1^k - \widehat{x}^k_{\text{end}}\big)$ identically. Discarding the shared-noise structure is therefore not a mild modelling simplification but the removal of the only mechanism by which the noise can be cancelled — and that structure is the stated contribution of the submission.

How to check. The exactness test of Section 7, Stage 1, run on all three estimators side by side with the exact displacement (49): (iii) returns x_0^k to solver tolerance, (ii) returns it up to the $\ker(G)$ scaling above, and (45) returns an estimate whose error is proportional to x_0^k with the coefficient tabulated above. Establishing first which of the two the released code executes is a prerequisite, since the figure caption and Equation (7) disagree.

Summary. Table 2 collects the comparison. Neither estimator in the submission is exact under the conditions in which the other is: the weighted least-squares fit is biased even with exact endpoints and no blur, because it discards the shared-noise structure that makes the system determined; and Equation (7) preserves that structure but discards the blur, so it is exact only at $G = I$. Equation (38) keeps both, reducing to Equation (7) when $G = I$ and to Prop. 7 as $\gamma \to 0$. Since the shared-noise structure is the stated contribution of the submission, replacing (45) by (38) also brings the implementation into agreement with the claim.

6.2 The anchor competes with the likelihood, and data consistency suffers

This is the hypothesis most likely to explain a poor measurement fit. In the submitted gradient the anchor term $\lambda_{\text{anc}}(\text{sg}(\widehat{x}_1^k) - x_1^k)$ acts on *every* direction, including those in which the measurement is fully informative. Consider a direction in which $A_k^\top A_k = 1$, for instance an observed pixel in inpainting. Setting the drift to zero and retaining the two dominant terms,

$$\underbrace{\tfrac{1}{\eta^2}\big(y - x_1^k\big)}_{\text{likelihood}} + \underbrace{\lambda_{\text{anc}}\big(\widehat{x}_1^k - x_1^k\big)}_{\text{anchor}} = 0 \qquad \Longrightarrow \qquad x_1^k = \alpha\, y + (1-\alpha)\,\widehat{x}_1^k, \quad \alpha = \frac{1/\eta^2}{1/\eta^2 + \lambda_{\text{anc}}}. \tag{48}$$

The fixed point does not depend on the preconditioner, since P_k is positive definite and $P_k g = 0$ if and only if $g = 0$. With the reported $\eta = 0.05$, so $1/\eta^2 = 400$:

λ_{anc}	150	200	300	350
α	0.73	0.67	0.57	0.53
displacement from y toward $\widehat{x}_1^k$	27%	33%	43%	47%

So even in directions where the data determine the answer to within η, the iteration is pulled between a quarter and a half of the way from y toward the prior's denoised estimate. The measurement residual at convergence is then not at the noise floor but at a fixed fraction of $\|\widehat{x}_1^k - y\|$, which is orders of magnitude larger.

How to check. Three measurements, in increasing order of effort.

1. Plot the normalized residual $\|A_k x_1^k - y\|/(\eta\sqrt{m})$ over the sweep and at termination. It should settle near 1. A value in the tens or hundreds confirms (48).
2. For inpainting, restrict to the *observed* pixels and histogram $|x_1^k - y|$ there. Under correct behaviour this concentrates at η; under (48) it concentrates near $(1-\alpha)|\widehat{x}_1^k - y|$.
3. Set $\lambda_{\text{anc}} = 0$ and repeat. The residual should fall toward 1. This is a one-line change and is the cleanest confirmation, since by Section 1.8 the restoring force the anchor supplies is already contained in the score.

We stress that removing the anchor is not expected to cost anything. The restoring force toward $\widehat{x}_1^k$ that it provides is already present in $(H_\tau^k)^\top \widehat{s}_{\text{flow}}$, with the weight $(H_\tau^k)^\top H_\tau^k/(\sigma_\tau^k)^2$ that Prop. 2 shows is the largest the curvature admits. Setting $\lambda_{\text{anc}} = 0$ therefore removes a term that is simultaneously redundant, in excess of the curvature bound, and by (48) actively opposed to the measurement. It is reasonable to expect performance to *improve* rather than degrade, and since it is a one-line change it should be the first experiment run.

6.3 The noise endpoint is effectively frozen

The Block-2 update contracts by $(1 - h_0/2)$ per iteration, so its relaxation time is $2/h_0$ steps. The reported values give

	h_0	relaxation time $2/h_0$	budget per stage $N_k S$
ImageNet tasks	10^{-3}	2000 steps	≈ 100 steps
fastMRI	10^{-2}	200 steps	≈ 100 steps

On the ImageNet settings the budget is about five per cent of a single e-folding: within a stage, x_0^k barely moves from the fresh draw it was initialized with. The joint refinement of (x_1^k, x_0^k) that the method advertises is then largely inactive, and the sampler behaves close to one that holds the noise endpoint fixed at a prior draw — which is what the methods it is compared against do. It is suggestive that fastMRI, the task with the larger h_0 and hence the only setting reaching half an e-folding, is also the task with the strongest reported result.

How to check. Record $\|x_0^{k(s)} - x_0^{k(0)}\|/\|x_0^{k(0)}\|$ across the inner iterations of a stage. If it remains near zero, h_0 is too small for the budget. Raising h_0 toward 10^{-1} and re-measuring is the corresponding experiment.

6.4 The clean-image step closes only part of the gap

With the derived preconditioner the Block-1 update contracts the distance to the conditional mean by $(1 - h_1/2)$ per iteration, so after S inner iterations a fraction $(1 - h_1/2)^S$ remains. The reported settings give $(1 - 0.05)^{15} \approx 0.46$ for $h_1 = 0.1, S = 15$ and $(1 - 0.1)^{10} \approx 0.35$ for $h_1 = 0.2, S = 10$: between a third and a half of the gap is still open when the sampler moves to the next τ, and the target has moved by then.

How to check. At a fixed (k, τ), compute the exact conditional mean $M_\tau^{-1} b_\tau$ by one additional CG solve, and record $\|x_1^{k(s)} - M_\tau^{-1} b_\tau\|$ across the inner iterations. This gives the fraction of the gap closed directly, and can be compared against $(1 - h_1/2)^S$.

6.5 An instrumentation checklist

All four hypotheses are resolved by logging six quantities per inner iteration, at negligible cost. We suggest recording them for one image before changing anything else.

Quantity	What it tells you
$\|A_k x_1^k - y\|/(\eta\sqrt{m})$	data consistency; should approach 1
$\|x_0^k\|/\sqrt{n}$	noise-endpoint scale; should approach 1
$\|x_0^{k(s)} - x_0^{k(0)}\|/\|x_0^{k(0)}\|$	whether Block 2 is moving at all
$\|x_1^{k(s)} - M_\tau^{-1} b_\tau\|$	fraction of the Block-1 gap closed
$\|\frac{1}{\eta^2} A_k^\top (y - A_k x_1^k)\|$, $\|(H_\tau^k)^\top \widehat{s}_{\text{flow}}\|$, $\|\lambda_{\text{anc}}(\widehat{x}_1^k - x_1^k)\|$ separately	relative weight of likelihood, score and anchor across (k, τ)
$\|x_\tau^k - H_\tau^k x_1^k - \sigma_\tau^k x_0^k\|$	bookkeeping assertion; must be zero

The third and fifth rows are the ones that discriminate between the hypotheses above. If the anchor term dominates the other two at large σ_τ^k, and the noise endpoint does not move, then Sections 6.2 and the preceding subsection together account for the behaviour without any appeal to the formulation itself.

7 Developing and testing Algorithm 2

The algorithm has two linear solves and one network call, and each can be validated on its own before any of them are combined. We describe a staged procedure, using inpainting as the running example because its forward operator is a diagonal mask and therefore contributes nothing that has to be debugged. Throughout, the recommendation is to begin at the *finest* scale $k = K-1$, where $A_k = A$ and the scale-adaptation U^{K-1-k} is the identity, and only afterwards to introduce the coarser scales.

7.1 Stage 0: the operators

Implement A_k, $A_k^\top$, G, H_τ^k, B_k and N_k as matrix-free routines. For inpainting A is a diagonal 0/1 mask, so $A^\top = A$ and $A^\top A = A$. Four tests, all cheap and all on random vectors:

1. *Adjoint test.* $\langle A_k v, w\rangle = \langle v, A_k^\top w\rangle$ to machine precision, and likewise for G and H_τ^k. This catches transpose and boundary errors, which are the usual source of silent failure.
2. *Symmetry.* $\|Gv - G^\top v\|$ and $\|H_\tau^k v - (H_\tau^k)^\top v\|$ negligible. Symmetry of G is assumed throughout Section 1 and used in every commutation argument.
3. *Spectral bound.* Power iteration on G should give $\lambda_{\max}(G) \le 1$, and $\langle v, Gv\rangle \ge 0$ for random v.
4. *Commutation.* $\|B_k H_\tau^k v - H_\tau^k B_k v\|$ negligible, and the same for N_k and H_τ^k. The elimination in Prop. 7 and the algebra of Cor. 8 both rest on this.

7.2 Stage 1: the score solve, in isolation

The operator is $N_k^2 + \gamma^2 (H_\tau^k)^2$, a polynomial in G and therefore symmetric positive definite. Verify positivity by $\langle v, (N_k^2 + \gamma^2 (H_\tau^k)^2) v\rangle > 0$, and on a small image (say 32×32) form the operator densely and compare a dense solve against the CG output.

The decisive test is the following, and it validates the algebra of Prop. 7, the operator implementations, and the solver simultaneously. Take a real image x_1^k, draw $x_0^k \sim \mathcal{N}(0, I)$, and form

$$x_\tau^k = H_\tau^k x_1^k + \sigma_\tau^k x_0^k, \qquad d_\tau^k = B_k x_1^k - (e_k - s_k) x_0^k, \tag{49}$$

the second from (3). Both are computable because x_1^k and x_0^k are known. Now solve (38) with $\gamma = 0$, using this exact d_τ^k in place of v_θ. The result must equal x_0^k to the solver tolerance, since Prop. 7 is an identity when the displacement is exact. Any discrepancy is a bug, not an approximation.

Repeating the test with $v_\theta(x_\tau^k, \tau, k)$ in place of d_τ^k then measures the network error directly: $\|\widehat{x}_0^k - x_0^k\|$ as a function of τ and k. This is how γ^2 is obtained, and Section 7.11 returns to it.

Two further checks make the output inspectable rather than only numeric.

Look at the implied clean image. From $\widehat{x}_0^k$ recover $\widehat{x}_1^k = (H_\tau^k)^{-1}(x_\tau^k - \sigma_\tau^k \widehat{x}_0^k)$ and display it. With the exact displacement it is the ground truth exactly. With v_θ it should look like a plausible denoising of x_τ^k: recognizably the right image, blurry at small τ and sharpening as $\tau \to 1$, and never noisier than x_τ^k itself. If it looks like noise, or is sharper than the ground truth, a scale convention or a time index is wrong, and no amount of tuning will repair it.

Split the error by frequency band. Project $\widehat{x}_1^k - x_1^k$ onto the range of G and onto $\ker(G)$ and plot the two errors separately against τ. This is the measurement that exposes the missing-blur error of Section 6.1: the two estimators agree in the low band and differ in the high band by the factor tabulated there.

7.3 Stage 1b: testing the regularization of Corollary 8

The tests above validate Prop. 7, the elimination. They say nothing about the ridge $\gamma^2(H_\tau^k)^2$ that Cor. 8 adds, which is a separate modelling assumption: that the velocity error is Gaussian, white, of variance γ^2, and independent of X_0^k. Four further checks address it. We note first that the infill test of Stage 2 does *not* test Cor. 8 either: it uses no network at all, by design, so that the linear algebra of the Block-1 solve can be validated without the score. The two stages isolate complementary halves.

(i) The $\gamma \to 0$ limit. With $\gamma = 0$, Cor. 8 reduces to Prop. 7, and the exactness test of Stage 1 applies unchanged. Also solve the unsquared system $N_k \widehat{x}_0^k = z_\tau^k$ directly and confirm the two agree; a discrepancy indicates a conditioning problem rather than an algebraic one, which is the subject of the next check.

(ii) Solve it as a least-squares problem, not a normal equation. Equation (38) is the normal equation of

$$\widehat{x}_0^k = \arg\min_x \ \left\|N_k x - z_\tau^k\right\|^2 + \gamma^2\left\|H_\tau^k x\right\|^2 \ = \ \arg\min_x \ \left\| \begin{bmatrix} N_k \\ \gamma H_\tau^k \end{bmatrix} x - \begin{bmatrix} z_\tau^k \\ 0 \end{bmatrix} \right\|^2, \tag{50}$$

so applying conjugate gradients to $N_k^2 + \gamma^2(H_\tau^k)^2$ squares the condition number. Running LSQR or CGLS on the stacked system of (50) instead sees only the square root of that, at the same cost per iteration. The test is to confirm the two produce the same answer, and to compare their iteration counts; if they differ appreciably, use the stacked form.

(iii) Calibration of γ^2: the decisive test. If the error model behind Cor. 8 is adequate, then the value of γ that minimizes the reconstruction error of $\widehat{x}_0^k$ should coincide with the velocity-error magnitude measured independently by (56). Both are available in a test harness, where x_0^k is known. Sweep γ, record $\|\widehat{x}_0^k - x_0^k\|$, and compare the minimizer against $\sqrt{\gamma^2}$ from (56). A synthetic check in which the velocity error is Gaussian of known standard deviation 0.30 gives

γ	0	0.10	0.20	0.25	**0.30**	0.40	0.50	1.00
RMSE of $\widehat{x}_0^k$	0.2367	0.2348	0.2298	0.2275	**0.2267**	0.2308	0.2433	0.3925

with the minimum at exactly the true value. Repeating this with the real network answers the question the submission left open. If the empirical optimum sits close to (56), the error model is adequate and γ^2 need never be tuned. If it sits far away, the model is misspecified — the velocity error is coloured, or correlated with X_0^k — and that should be reported rather than absorbed by tuning γ.

(iv) Shrinkage monotonicity. Cor. 8 is a shrinkage estimator, so $\|\widehat{x}_0^k\|/\sqrt{n}$ should decrease monotonically in γ, approaching 1 as $\gamma \to 0$ when the network is accurate, and 0 as $\gamma \to \infty$. Non-monotonicity indicates a sign or scaling error in the assembly of z_τ^k.

(v) The counterpart of the infill test. The infill test asks whether the linear algebra propagates prior information into the unobserved region. Its counterpart here asks how much the prior knows in the first place, and uses no measurement. Build x_τ^k from the ground truth, run Cor. 8 with the real network, form $\widehat{x}_1^k = (H_\tau^k)^{-1}(x_\tau^k - \sigma_\tau^k \widehat{x}_0^k)$, and compare it against x_1^k as an image and by

frequency band, sweeping τ and k. The error should fall as $\tau \to 1$ and should be larger at coarse k. This curve is the prior's information content at each point of the schedule, and it upper-bounds what any sampler built on it can recover: if $\widehat{x}_1^k$ is poor at a given (k, τ), no choice of step sizes will produce a good reconstruction there.

7.4 Stage 1c: the correct reference, and what the images should look like

Building x_τ^k from the ground truth and solving (38) is the right experiment, but the reference to compare against depends on what is fed in as the displacement, and getting this wrong produces a false alarm. The estimator targets $\bar{x}_0^k = \mathbb{E}[X_0^k \mid x_\tau^k]$, a *conditional mean*, and not the particular x_0^k that was drawn. The two are different random variables.

(A) Realized displacement: the reference is the drawn x_0^k. Feed $d = B_k x_1^k - (e_k - s_k)x_0^k$, computed from the same x_1^k and x_0^k used to build x_τ^k. Then (1) is a determined 2×2 system in those two vectors, and the solve returns the drawn x_0^k *exactly*, to solver tolerance. This is the algebra-and-solver test of Stage 1, and it is the only setting in which an exact match to x_0^k should be expected.

(B) Conditional-mean displacement: the reference is $\mathbb{E}[X_0^k \mid x_\tau^k]$. This is what v_θ approximates, and the output is then $\mathbb{E}[X_0^k \mid x_\tau^k]$, which does *not* equal x_0^k. Taking a Gaussian prior Σ makes both sides computable: with everything diagonal in the Fourier basis, $\mathbb{E}[X_0^k \mid x_\tau^k] = \sigma_\tau^k (H_\tau^k \Sigma H_\tau^k + (\sigma_\tau^k)^2)^{-1} x_\tau^k$ and $\mathbb{E}[X_1^k \mid x_\tau^k] = \Sigma H_\tau^k (H_\tau^k \Sigma H_\tau^k + (\sigma_\tau^k)^2)^{-1} x_\tau^k$. Feeding the exact conditional-mean displacement into (38) with $\gamma = 0$ and comparing gives

τ	σ_τ^k	$\lVert\widehat{x}_0^k - \mathbb{E}[X_0^k \mid x_\tau^k]\rVert$	$\lVert\widehat{x}_0^k - x_0^k\rVert$	$\mathrm{corr}(\widehat{x}_0^k, x_0^k)$	$\lVert\widehat{x}_1^k - \mathbb{E}[X_1^k \mid x_\tau^k]\rVert$
0.1	0.725	$8 \cdot 10^{-15}$	3.79	0.993	$1 \cdot 10^{-13}$
0.3	0.675	$8 \cdot 10^{-15}$	5.08	0.988	$3 \cdot 10^{-14}$
0.5	0.625	$7 \cdot 10^{-15}$	6.01	0.982	$2 \cdot 10^{-14}$
0.7	0.575	$7 \cdot 10^{-15}$	8.12	0.968	$1 \cdot 10^{-14}$
0.9	0.525	$6 \cdot 10^{-15}$	9.88	0.948	$1 \cdot 10^{-14}$

The agreement with the conditional mean is at machine precision at every τ, while the discrepancy from the drawn x_0^k is large and grows as σ_τ^k decreases. Both columns are correct behaviour: as σ_τ^k shrinks, x_τ^k is dominated by $H_\tau^k x_1^k$ and carries progressively less information about x_0^k, so the conditional mean shrinks away from the realized value. Reporting the second column as an error would be a misdiagnosis.

(C) Subtracting to recover the image. The subtraction is exactly the right thing to do, and it has a precise meaning. Taking conditional expectations in (2) gives $x_\tau^k = H_\tau^k \bar{x}_1^k + \sigma_\tau^k \bar{x}_0^k$, so

$$\widehat{x}_1^k := (H_\tau^k)^{-1}\big(x_\tau^k - \sigma_\tau^k\, \widehat{x}_0^k\big) \tag{51}$$

is the implied estimate of $\mathbb{E}[X_1^k \mid x_\tau^k]$, that is, the minimum mean-squared-error denoising of the interpolant. What it should look like:

- Under test (A) it is the ground truth exactly. Displaying it is the fastest way to confirm the whole path.

- Under (B) or with a real network it is a *denoised*, not a perfect, image: recognizably the right scene, smooth and washed out at large σ_τ^k, sharpening as $\tau \to 1$. It should never be noisier than x_τ^k, and never sharper than x_1^k. Expecting $\widehat{x}_1^k = x_1^k$ at small τ is the same misdiagnosis as above.
- The residual $x_1^k - \widehat{x}_1^k$ is the detail the prior could not recover at that noise level. Displayed across the schedule it should shrink monotonically, and it localizes where the prior is weak.

A free consistency assertion. Subtracting (51) from (2) gives the identity

$$H_\tau^k\big(\widehat{x}_1^k - x_1^k\big) \;=\; \sigma_\tau^k\big(x_0^k - \widehat{x}_0^k\big), \tag{52}$$

so the image-domain and noise-domain errors are the same quantity in two coordinate systems. Equation (52) holds by construction and must be satisfied to machine precision; checking it costs nothing and catches sign and scaling errors in (51) immediately.

7.5 Stage 2: the Block-1 draw, in isolation

The operator is M_τ of (33). Check symmetry and positivity as above. For $\tau > 0$ a lower bound is available in closed form: since $G \succeq 0$,

$$H_\tau^k = (1-\tau)s_k G + \tau e_k I \;\succeq\; \tau e_k I \qquad \Longrightarrow \qquad M_\tau \;\succeq\; \frac{(\tau e_k)^2}{(\sigma_\tau^k)^2}\, I, \tag{53}$$

so the smallest eigenvalue is known and the CG iteration count can be predicted.

Two functional tests.

The mean, and whether the prior is filling the holes. This is the most informative test of the Block-1 solve, and it requires no network at all. Take the ground-truth image x_1^k, draw $x_0^k \sim \mathcal{N}(0, I)$, and build a *legitimate* interpolant from them, $x_\tau^k = H_\tau^k x_1^k + \sigma_\tau^k x_0^k$. Form y from the same x_1^k through the mask. Now set $\xi_y = \xi_h = 0$ in (34) and solve, so that the output is the conditional mean $M_\tau^{-1} b_\tau$. Because x_τ^k was constructed from the true image, the answer is predictable in both regions:

$$\underbrace{\big\|x_1^k - \text{output}\big\|_{\text{observed}} \;\approx\; \eta,}_{\text{the data pins these pixels}} \qquad \underbrace{\big\|x_1^k - \text{output}\big\|_{\text{missing}} \;\approx\; \sigma_\tau^k \,\big\|(H_\tau^k)^{-1} x_0^k\big\|}_{\text{only the interpolant informs these}}. \tag{54}$$

On the missing pixels $A_k^\top A_k$ vanishes, so the solve returns $(H_\tau^k)^{-1} x_\tau^k = x_1^k + \sigma_\tau^k (H_\tau^k)^{-1} x_0^k$ there: the true image plus noise at exactly the level of the blurred prior (9). The test therefore shows directly that the prior is filling the missing region, and shows it quantitatively.

Sweeping τ from near 0 to near 1 makes this sharper still: the observed-region error should stay flat at η, while the missing-region error should fall like σ_τ^k. A numerical example on a one-dimensional signal of length 256 with a 50% mask, a Gaussian G, $s_k = 0.25$, $e_k = 0.5$ and $\eta = 0.05$ gives

τ	σ_τ^k	error, observed	error, missing
0.05	0.737	0.044	17.1
0.25	0.688	0.051	4.24
0.50	0.625	0.050	2.44
0.75	0.562	0.043	1.36
0.95	0.512	0.054	0.997

matching (54) to within ten per cent at every τ. Two failure modes are caught immediately: an observed-region error much larger than η means the likelihood term or its scaling is wrong, and a missing-region error that does not fall with σ_τ^k means H_τ^k, its adjoint, or the right-hand side is wrong.

Only after this passes is it worth comparing a dense solve of (33) on a small problem, which checks the CG implementation rather than the formulation.

The covariance. This is the test of Lem. 5, and it is worth doing carefully, since an error in the noise construction produces samples with the right mean and the wrong spread, which is invisible in reconstruction metrics. Hold x_τ^k and y fixed, draw N independent x_1^k by the full right-hand side (34), and for a few fixed random vectors u compare

$$\underbrace{\widehat{\mathrm{Var}}\big(\langle u, x_1^k\rangle\big)}_{\text{empirical, over the } N \text{ draws}} \qquad \text{against} \qquad \underbrace{\langle u, M_\tau^{-1}u\rangle}_{\text{one extra CG solve}} \quad . \tag{55}$$

For inpainting there is a useful special case: on the unobserved pixels $A_k^\top A_k$ vanishes, so M_τ reduces to $(H_\tau^k)^\top H_\tau^k/(\sigma_\tau^k)^2$ there and the predicted variance is $(\sigma_\tau^k)^2(H_\tau^k)^{-2}$, available without a solve.

7.6 Stage 3: the Block-2 step, in isolation

Fix x_1^k and supply the exact score. The chain (36) should have stationary density proportional to $p_\tau^k(H_\tau^k x_1^k + \sigma_\tau^k x_0^k)\mathcal{N}(x_0^k; 0, I)$. Two checks are available without knowing p_τ^k. Replacing p_τ^k by a Gaussian of known covariance makes the stationary law closed-form, so the empirical mean and covariance can be compared directly. And in that Gaussian case the discretization bias is known: the stationary variance is inflated by $1/(1 - h_0/4)$, which can be confirmed by sweeping h_0.

7.7 Stage 4: one grid point, both blocks

Fix k and τ, disable the annealing, and run the two blocks for many sweeps. The chain should equilibrate, and three diagnostics are available at every iteration:

1. *Measurement residual.* $\|A_k x_1^k - y\|/(\eta\sqrt{m})$ should settle near 1: the reconstruction should agree with the data to the noise floor, not beyond it. A value far above 1 means data consistency is not being enforced; far below 1 means the measurement is being overfitted.
2. *Noise-endpoint scale.* $\|x_0^k\|/\sqrt{n}$ should settle near 1, since the target carries $\mathcal{N}(x_0^k; 0, I)$.
3. *Interpolant consistency.* $\|x_\tau^k - H_\tau^k x_1^k - \sigma_\tau^k x_0^k\|$ must be zero to machine precision at all times. This is not a convergence diagnostic but a correctness assertion on the bookkeeping of the restore line, and it should be left in as an assertion during development.

7.8 Stage 4b: the oracle-score run

Before any Gaussian surrogate, one integration test separates sampler bugs from score bugs completely. Replace $v_\theta(x_\tau^k, \tau, k)$ throughout by the exact displacement d_τ^k of (49), computed from the ground-truth x_1^k and the current x_0^k — an oracle no real run has, but one that is available in a test harness. Every approximation in the sampler except the blurring (9) is then removed, so the reconstruction should be excellent: at the finest scale, essentially exact in the observed region and limited only by σ_τ^k in the unobserved one.

If the oracle run reconstructs well and the real run does not, the sampler is correct and the problem is in the network or in the endpoint estimator built from it. If the oracle run also fails, the problem is in the sampler, and none of the score-side work will help. We recommend running this before any hyperparameter search, since it partitions the search space in a single experiment.

7.9 Stage 5: end-to-end with a prior of known form

The strongest correctness test replaces p_τ^k by a Gaussian, $\mathcal{N}(0, \Sigma)$ with Σ chosen diagonal in the same basis as G. Every conditional is then closed-form, the joint (16) is Gaussian, and its exact mean and covariance can be computed. Running the full sweep at fixed (k, τ) and comparing the sample mean and a few projections of the sample covariance against the analytic values tests the entire construction at once, including the interaction between the two blocks. This is the test that would catch an invariance error that the per-block tests miss.

7.10 Stage 6: one scale, then the cascade

Run a single stage k in isolation, initialized at the downsampled ground truth plus noise at the stage's own level. The outputs to record are x_1^k at that scale, the three diagnostics of Stage 4, and the CG iteration counts. Correctness at this level means: PSNR of x_1^k against $D^{K-1-k}x_1$ is comparable to what an oracle reconstruction at that resolution would give, and the measurement residual sits at the noise floor.

Only then assemble the cascade. The transition to be checked is the one line $x_1^{k+1} \leftarrow U^{(1)}(x_1^k)$ with $x_0^{k+1} \sim \mathcal{N}(0, I)$: compare the resulting initial x_τ at stage $k+1$ against the distribution the pretrained model was trained on at that point. A mismatch here produces a quality drop at each stage boundary that is easily read off a plot of PSNR against global time.

7.11 Hyperparameters: what to fix, what to measure, what to tune

The quantities separate into three groups, and only the last should be searched over.

Fixed. The ridge ϵ, discussed below, and the CG iteration count L, which is set by a residual tolerance rather than tuned; report the tolerance and the resulting mean iteration count.

Measured. The velocity-error variance γ^2. It is estimated, not searched, by the procedure of Stage 1: on held-out data where x_1^k and x_0^k are known, form the exact d_τ^k of (49) and set

$$\gamma^2 \approx \frac{1}{n}\,\mathbb{E}\big\|v_\theta(x_\tau^k, \tau, k) - d_\tau^k\big\|^2, \tag{56}$$

tabulated per (k, τ). Reporting (56) also answers the question of how accurate the endpoint estimate is, which the submitted version left open.

Tuned. For Algorithm 2, the single step size h_0, and then S as a compute budget. We suggest sweeping h_0 on a small validation set at fixed S, and then raising S until the metrics saturate. For Algorithm 1 the step size $h_1 \in (0, 2)$ is swept as well; Prop. 2 guarantees stability over that whole interval, so the sweep needs no safeguard.

7.12 Is γ a hyperparameter?

Not in the sense that a step size is. It is a nuisance parameter of a stated error model, $v_\theta = d_\tau^k + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \gamma^2 I)$, and it describes a measurable property of the trained network rather than a free choice about the sampler. It should therefore be *measured*, by (56) on held-out data, and reported — not searched over to optimize a reconstruction metric. Three things make this practical rather than aspirational.

It is verifiable. The calibration test of Section 7.3(iii) checks the claim directly: the γ minimizing $\|\widehat{x}_0^k - x_0^k\|$ should coincide with the value (56) returns. If it does, γ^2 has been measured and the count in Table 3 stands. If it does not, the error model is misspecified — most likely the velocity error is correlated across pixels rather than white — and honesty then requires either generalizing the model or reclassifying γ as tuned and counting it.

It need not be accurate. The estimator is insensitive to γ over a wide range. In the synthetic setting of Section 7.3, with the true value $\gamma_\star$:

$\gamma/\gamma_\star$	0	0.5	0.83	1.0	1.17	1.5	2.0
excess RMSE over the optimum	4.2%	2.3%	0.1%	—	0.5%	4.4%	17%

A factor of two in γ costs under five per cent, and only gross misspecification is expensive. A parameter with this profile does not reward tuning, which is precisely why measuring it once is adequate.

It is one number per grid point, not one number. The velocity error depends on the noise level, so (56) should be tabulated per (k, τ) and stored with the model. If the error turns out to be coloured, the natural generalization replaces $\gamma^2 I$ by a covariance Σ_ε, and (38) becomes $\big[N_k(H_\tau^k \Sigma_\varepsilon H_\tau^k)^{-1} N_k + I\big]\widehat{x}_0^k = N_k(H_\tau^k \Sigma_\varepsilon H_\tau^k)^{-1} z_\tau^k$; if Σ_ε is diagonal in the same basis as G, this is still one measurement per frequency band and still not a search.

7.13 Is ϵ needed at every step?

No. By (53), $M_\tau \succeq (\tau e_k)^2 (\sigma_\tau^k)^{-2} I$, so M_τ is already positive definite for every $\tau > 0$ and no ridge is required there. The one degenerate point is $\tau = 0$, where $H_0^k = s_k G$ is rank-deficient and M_τ is singular on $\ker(G) \cap \ker(A_k)$; this is the situation of Prop. 4.

Two remarks follow. First, the cleanest remedy is not to add anything but to start the time grid at the first $\tau > 0$. At $\tau = 0$ the interpolant is x_{start}^k, which carries no information about x_1^k in $\ker(G)$, so nothing is lost by omitting that grid point.

Second, if ϵ is retained, it should be treated as a numerical safeguard and not as a hyperparameter. The bound (53) degrades as $\tau \to 0$, so at small τ the system is poorly conditioned even when invertible and CG converges slowly; a small ϵ improves that at negligible bias. Set it relative to the scale of the operator, for instance $\epsilon = 10^{-6}\,\|M_\tau\|$, fix it once, and report it. It should *not* be tuned: if performance depends appreciably on ϵ, then ϵ is doing modelling work rather than conditioning work, and that dependence should be reported rather than optimized away.

8 Relation to minorize–maximize mode estimation

The blurred prior of Assumption (A) is the same object that appears in minorize–maximize (MM) schemes for diffusion priors, and the comparison clarifies which parts of that machinery are needed here and which are not.

The same $\sqrt{2}$. An MM scheme of that kind targets $\log p(y \mid x) + \log p_{\sqrt{2}\sigma_t}(x)$, where the $\sqrt{2}$ arises because the density p_{σ_t} is evaluated at a point that has itself been perturbed by a further σ_t of noise, and two independent Gaussian perturbations compose to one of scale $\sqrt{2}\sigma_t$. That is the mechanism of Lemma 1(b): p_τ^k already contains one σ_τ^k, and integrating against $\mathcal{N}(x_0^k; 0, I)$ supplies the second. The blurring is therefore the same, with H_τ^k in place of the identity.

Why a Jensen bound is not needed here. The two formulations part company in what they do with the expectation. An MM scheme must ascend a *deterministic* objective, so the expectation over the perturbation has to be absorbed into that objective; moving it inside the logarithm, $\mathbb{E}[\log p_{\sigma_t}]$ in place of $\log \mathbb{E}[p_{\sigma_t}]$, is what Jensen's inequality supplies, together with a one-sided bound and a gap equal to a Kullback–Leibler divergence. The bound is required because monotone ascent needs a minorizer, and because the resulting expectation forces the network to be evaluated at noisy inputs, where its score estimate is reliable.

Neither requirement arises for a sampler. First, the perturbation is not integrated out at all: the noise endpoint x_0^k is retained as a state variable, and the chain marginalizes it. What appears in (17) is therefore $\log \mathbb{E}[p_\tau^k]$ itself, the exact blurred prior, rather than the Jensen lower bound $\mathbb{E}[\log p_\tau^k]$. Second, a Gibbs sweep needs only the gradient of the target, never a one-sided bound on it, so no minorizer is constructed. Third, the network is evaluated at $x_\tau^k = H_\tau^k x_1^k + \sigma_\tau^k x_0^k$ at every iteration, which is by construction a point of the kind the model was trained on; the role played by an explicit renoising step is here played by the state variable x_0^k.

What the two do share is the curvature bound. Proposition 2 is the statement that makes a step size or a preconditioner selectable without a Lipschitz constant, and it is used in both settings for the same reason.

Averaging over noise realizations. An MM scheme must estimate its expectation, and does so with S renoised draws per step; S then controls both the Monte Carlo variance and the reproducibility of the output. No corresponding average is called for here. The quantity $(H_\tau^k)^\top \nabla \log p_\tau^k(x_\tau^k)$ used by both algorithms is not an estimate of an expectation: it is the exact gradient of the joint log-density with respect to x_1^k at the current state, with x_0^k held at its current value. Replacing it by an average over fresh draws of x_0^k would substitute a different vector field, and neither block would leave (16) invariant.

It matters which average is meant, since the two differ by exactly the Jensen gap. Averaging the network score over $x_0^k \sim \mathcal{N}(0, I)$, that is over draws from the *prior*, gives the gradient of the renoise–denoise regularizer, which is the lower bound. Averaging over $x_0^k \sim q(\cdot \mid x_1^k)$, draws from the *conditional*, gives the exact score of the blurred prior by (14). An MM scheme takes the first. The sampler needs neither: x_0^k is carried in the state, and Block 2 visits $q(\cdot \mid x_1^k)$ over successive sweeps, so the role played by S in an MM scheme is played here by the number of inner iterations. If more averaging is wanted, that is the quantity to increase, not a per-step average.

The S-sample average of an MM scheme is also motivated by reproducibility, since a deterministic reconstruction is the goal there. For a sampler the dependence on the random seed is the diversity being sought, so that motivation does not carry over. One case remains in which a small average might help in practice: it damps the variance of a *network* error, as opposed to a Monte Carlo error, which may matter at coarse stages where σ_τ^k is large and the endpoint estimate is least informative. Such an average biases the sampler away from (16), so it should be reported as a heuristic and compared against simply increasing S, which smooths in the same direction without leaving the target.

A mode-estimation variant. The parallel suggests a variant worth investigating. The Block-1 conditional (30) is Gaussian, and Algorithm 2 draws from it by solving $M_\tau x_1^k = \widetilde{b}_\tau$. Replacing the perturbed right-hand side $\widetilde{b}_\tau$ by b_τ solves the same system for the *maximizer* of that conditional rather than a sample from it. The result is a deterministic reconstruction obtained at identical cost, and the curvature bound of Proposition 2 is exactly the ingredient an MM analysis of such an iteration requires. We leave the convergence analysis of that variant, and its comparison against the sampler on reproducibility and runtime, to future work.

A Proofs

Proof of Lemma 1

It is worth saying first what (12) is doing. Equation (11) maps the clean image X to the interpolant by scaling with H^k_τ and adding noise $\sigma^k_\tau Z_1$. To recover X one would subtract that noise and apply $(H^k_\tau)^{-1}$. The noise Z_1 is not known, so (12) subtracts an *independent* copy $\sigma^k_\tau X^k_0$ in its place. It is therefore an attempted inversion of (11), and what it returns is not X but X with further noise added. Steps 1 and 2 compute how much.

Part (a): the law of X^k_1.

Step 1. Substitute (11) into (12) and expand. The factor $(H^k_\tau)^{-1}$ cancels the H^k_τ multiplying X:

$$X^k_1 = (H^k_\tau)^{-1}\big(\underbrace{H^k_\tau X + \sigma^k_\tau Z_1}_{X^k_\tau} - \sigma^k_\tau X^k_0\big) = X + \sigma^k_\tau\,(H^k_\tau)^{-1}\big(Z_1 - X^k_0\big). \tag{57}$$

The clean image is recovered exactly; what remains is the residual of the two noises.

Step 2. Combine them. Both Z_1 and X^k_0 are standard and independent, so their difference has covariance $I + I = 2I$; that is, $Z_1 - X^k_0 = \sqrt{2}\,Z$ in law with $Z \sim \mathcal{N}(0, I)$. Substituting into (57),

$$X^k_1 \;=\; X + \sqrt{2}\,\sigma^k_\tau\,(H^k_\tau)^{-1} Z, \tag{58}$$

whose density is (9). This proves (a). The factor $\sqrt{2}$ is now explicit: one unit of noise is carried inside X^k_τ by (11), the second is the X^k_0 subtracted in (12), and because the two are independent their variances add.

Part (b): the conditional density.

Step 3. Condition on $X^k_0 = x^k_0$ in (12). With x^k_0 fixed, X^k_1 is an affine function of X^k_τ:

$$X^k_1 \;=\; \underbrace{(H^k_\tau)^{-1}}_{=:\,M} X^k_\tau \;+\; \underbrace{\big(-\sigma^k_\tau (H^k_\tau)^{-1} x^k_0\big)}_{=:\,c}. \tag{59}$$

Step 4. Recall the change-of-variables rule for densities: if U has density p_U and $V = MU + c$ with M invertible, then

$$p_V(v) \;=\; p_U\big(M^{-1}(v - c)\big)\,|\det M|^{-1}. \tag{60}$$

Apply (60) with $U = X^k_\tau$, whose density is p^k_τ, and with M and c read off from (59). The two ingredients it requires are

$$M^{-1}(x^k_1 - c) = H^k_\tau\Big(x^k_1 + \sigma^k_\tau (H^k_\tau)^{-1} x^k_0\Big) = H^k_\tau x^k_1 + \sigma^k_\tau x^k_0, \qquad |\det M|^{-1} = |\det (H^k_\tau)^{-1}|^{-1} = |\det H^k_\tau|. \tag{61}$$

Step 5. Substituting (61) into (60) gives $p(x^k_1 \mid x^k_0) = p^k_\tau\big(H^k_\tau x^k_1 + \sigma^k_\tau x^k_0\big)\,|\det H^k_\tau|$, which is (13).

Proof of Proposition 2

From (19), $\nabla^2_{x^k_1}(-\log \pi) = \frac{1}{\eta^2} A_k^\top A_k - (H^k_\tau)^\top\big[\nabla^2 \log p^k_\tau\big] H^k_\tau$ by the chain rule. It therefore suffices to bound $\nabla^2 \log p^k_\tau$ from below. Tweedie's identity for the linear-Gaussian interpolant (2) gives $\nabla \log p^k_\tau(x^k_\tau) = \big(H^k_\tau \mathbb{E}[X^k_1 \mid x^k_\tau] - x^k_\tau\big)/(\sigma^k_\tau)^2$, and differentiating once more,

$$\nabla^2 \log p^k_\tau(x^k_\tau) = \frac{1}{(\sigma^k_\tau)^2}\Big(H^k_\tau \frac{\partial \mathbb{E}[X^k_1 \mid x^k_\tau]}{\partial x^k_\tau} - I\Big), \qquad \frac{\partial \mathbb{E}[X^k_1 \mid x^k_\tau]}{\partial x^k_\tau} = \frac{\mathrm{Cov}(X^k_1 \mid x^k_\tau)\,(H^k_\tau)^\top}{(\sigma^k_\tau)^2},$$

the second identity being the standard sensitivity of a posterior mean in a linear-Gaussian channel. Since $\operatorname{Cov}(X_1^k \mid x_\tau^k) \succeq 0$, the matrix $H_\tau^k \operatorname{Cov}(X_1^k \mid x_\tau^k)(H_\tau^k)^\top$ is positive semidefinite and $\nabla^2 \log p_\tau^k \succeq -(\sigma_\tau^k)^{-2} I$. Hence $-(H_\tau^k)^\top[\nabla^2 \log p_\tau^k] H_\tau^k \preceq (H_\tau^k)^\top H_\tau^k/(\sigma_\tau^k)^2$, which gives (24).

Proof of Proposition 3

Holding x_τ^k fixed in (30), the third factor is constant and the terms depending on x_1^k are $-\|A_k x_1^k - y\|^2/2\eta^2 - \|x_\tau^k - H_\tau^k x_1^k\|^2/2(\sigma_\tau^k)^2$. Expanding both squares and discarding constants,

$$\log \pi(x_1^k \mid x_\tau^k, y) = -\tfrac{1}{2}\, {x_1^k}^\top \Big(\underbrace{\tfrac{1}{\eta^2} A_k^\top A_k}_{\text{from } p(y \mid x_1^k)} + \underbrace{\tfrac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top H_\tau^k}_{\text{from the interpolant}} \Big) x_1^k + {x_1^k}^\top \Big(\underbrace{\tfrac{1}{\eta^2} A_k^\top y}_{\text{from } p(y \mid x_1^k)} + \underbrace{\tfrac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top x_\tau^k}_{\text{from the interpolant}} \Big) + \text{const}, \tag{62}$$

that is, $-\frac{1}{2}{x_1^k}^\top M_\tau x_1^k + {x_1^k}^\top b_\tau + \text{const}$. Completing the square gives $-\frac{1}{2}(x_1^k - M_\tau^{-1} b_\tau)^\top M_\tau (x_1^k - M_\tau^{-1} b_\tau) + \text{const}$.

Proof of Proposition 4

If $u \in \mathcal{K}_\tau$ then $A_k(x_1^k + u) = A_k x_1^k$ and $H_\tau^k(x_1^k + u) + \sigma_\tau^k x_0^k = x_\tau^k$, so all three factors of (16) are unchanged. The density is constant along u and its integral diverges.

Proof of Lemma 5

The vector ζ is Gaussian with mean zero, and by independence of ξ_y and ξ_h,

$$\operatorname{Cov}(\zeta) = \frac{1}{\eta^2} A_k^\top A_k + \frac{1}{(\sigma_\tau^k)^2} (H_\tau^k)^\top H_\tau^k = M_\tau.$$

Hence $\widetilde{b}_\tau \sim \mathcal{N}(b_\tau, M_\tau)$, and applying M_τ^{-1} gives mean $M_\tau^{-1} b_\tau$ and covariance $M_\tau^{-1} M_\tau M_\tau^{-1} = M_\tau^{-1}$.

Proof of Proposition 6

Tweedie's identity for the linear-Gaussian interpolant (2) gives $\nabla \log p_\tau^k(x_\tau^k) = \big(H_\tau^k \mathbb{E}[X_1^k \mid x_\tau^k] - x_\tau^k\big)/(\sigma_\tau^k)^2$. Taking conditional expectations in $x_\tau^k = H_\tau^k x_1^k + \sigma_\tau^k x_0^k$, and using that x_τ^k is the conditioning variable, gives $H_\tau^k \mathbb{E}[X_1^k \mid x_\tau^k] - x_\tau^k = -\sigma_\tau^k\, \mathbb{E}[X_0^k \mid x_\tau^k]$. Divide by $(\sigma_\tau^k)^2$.

Proof of Proposition 7

Step 1 (conditioning). Both (2) and (3) are linear in (x_1^k, x_0^k). Conditioning them on $X_\tau^k = x_\tau^k$ and using linearity of the conditional expectation gives two linear equations in the two conditional means, the left-hand side of the first being known because x_τ^k is the conditioning variable:

$$\underbrace{x_\tau^k}_{\text{known}} = H_\tau^k\, \bar{x}_1^k + \sigma_\tau^k\, \bar{x}_0^k, \tag{63}$$

$$\underbrace{d_\tau^k}_{\text{estimated by } v_\theta} = B_k\, \bar{x}_1^k - (e_k - s_k)\, \bar{x}_0^k. \tag{64}$$

Step 2 (elimination of $\bar{x}_1^k$). B_k and H_τ^k are polynomials in G and commute, so $B_k H_\tau^k \bar{x}_1^k = H_\tau^k B_k \bar{x}_1^k$ and the combination $B_k \cdot$ (63) $- H_\tau^k \cdot$ (64) cancels $\bar{x}_1^k$:

$$B_k\, x_\tau^k - H_\tau^k\, d_\tau^k = \big[\sigma_\tau^k B_k + (e_k - s_k) H_\tau^k\big]\, \bar{x}_0^k. \tag{65}$$

Step 3 (the bracket). Substituting $B_k = e_k I - s_k G$ and $H_\tau^k = (1-\tau)s_k G + \tau e_k I$ and collecting the coefficients of I and G,

$$\sigma_\tau^k B_k + (e_k - s_k)H_\tau^k = \underbrace{e_k\big[\sigma_\tau^k + \tau(e_k - s_k)\big]}_{=\ e_k(1-s_k)}\ I\ -\ \underbrace{s_k\big[\sigma_\tau^k - (1-\tau)(e_k - s_k)\big]}_{=\ s_k(1-e_k)}\ G\ =:\ N_k, \tag{66}$$

both underbraced identities following from $\sigma_\tau^k = (1-\tau)(1-s_k) + \tau(1-e_k)$:

$$\sigma_\tau^k + \tau(e_k - s_k) = (1-\tau)(1-s_k) + \tau(1-s_k) = 1 - s_k, \qquad \sigma_\tau^k - (1-\tau)(e_k - s_k) = 1 - e_k.$$

Proof of Corollary 8

G, B_k, H_τ^k and N_k are symmetric and commute, being polynomials in G. The posterior mean minimizes

$$\tfrac{1}{2}\big(z_\tau^k - N_k x\big)^\top \big(\gamma^2 (H_\tau^k)^2\big)^{-1}\big(z_\tau^k - N_k x\big) + \tfrac{1}{2}\|x\|^2,$$

which is $-\log$ of $\underbrace{p(z_\tau^k \mid x_0^k)}_{\mathcal{N}(z_\tau^k;\, N_k x_0^k,\, \gamma^2 (H_\tau^k)^2)} \cdot \underbrace{p(x_0^k)}_{\mathcal{N}(x_0^k;0,I)} \propto \underbrace{p(x_0^k \mid z_\tau^k)}_{\text{posterior of } X_0^k}$ up to a constant. Its stationarity condition is $\big[N_k(\gamma^2(H_\tau^k)^2)^{-1}N_k + I\big]\widehat{x}_0^k = N_k(\gamma^2(H_\tau^k)^2)^{-1}z_\tau^k$; multiplying on the left by $\gamma^2(H_\tau^k)^2$ and using commutativity gives (38).